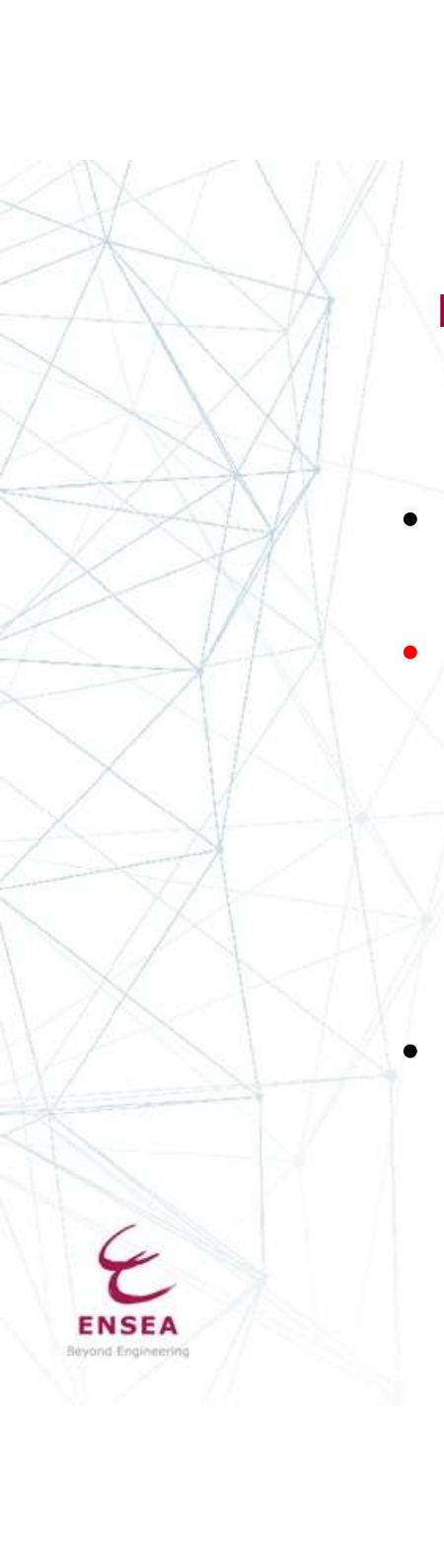




Power Electronics

alexis.martin@ensea.fr
Desk D216





Layout

- Non isolated choppers
- **Switch mode power supplies**
 - Introduction
 - Flyback
 - Forward
 - Push-Pull
- Power components

Introduction

Flyback

Forward

Push-Pull

Introduction

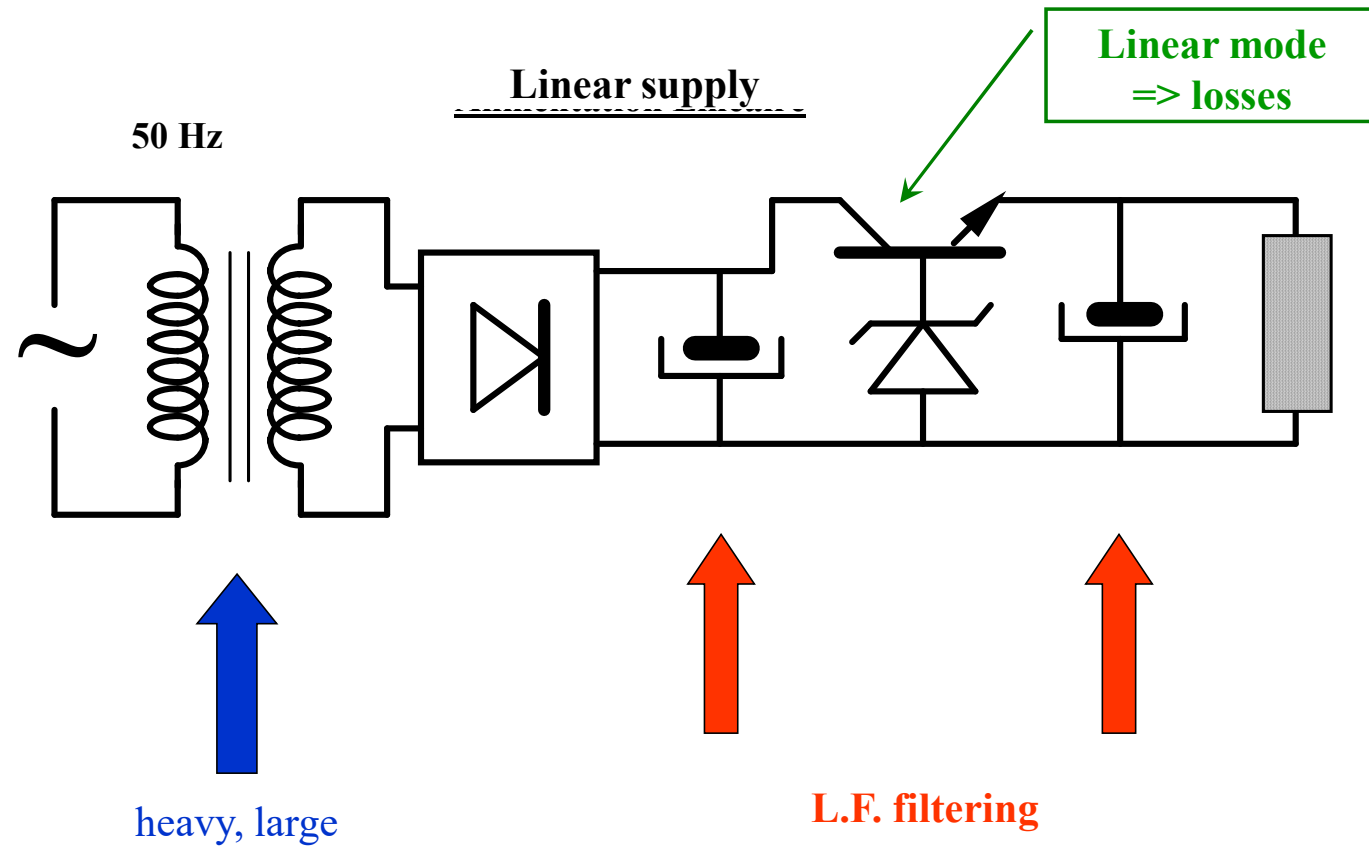
Introduction

Flyback

Forward

Push-Pull

Linear isolated power supply principle



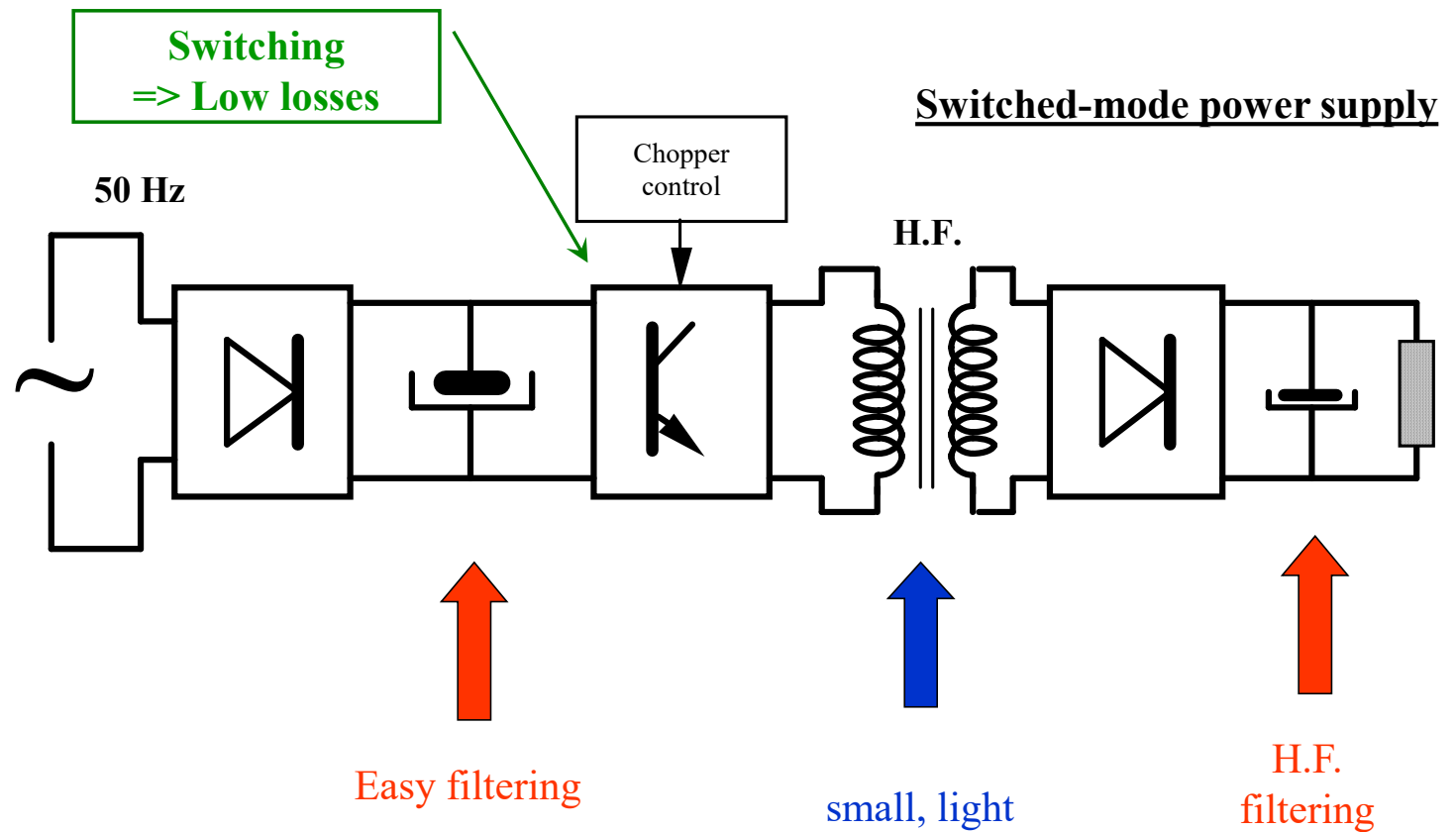
Introduction

Flyback

Forward

Push-Pull

Isolated switched-mode power supply principle



Introduction

Flyback

Forward

Push-Pull

Comparison

Element	LINEAR	Switching
Transformer	Large (50 Hz)	Small (H.F.)
Transistor	Linear (losses)	Switching
Output filter	Large (50 Hz)	Simple (H.F.)
Command	autonomous, simple	control, complex
Efficiency	40 % < < 50 %	70 % < < 90 %
Control	0,1 %	0,5 %

Introduction

Flyback

Forward

Push-Pull

Switched-mode power supply nomenclature

- ***Flyback** : energy storage converter*
- ***Forward** : direct converter*
- ***Push-Pull** : symmetric converter
(full bridge or half bridge)*

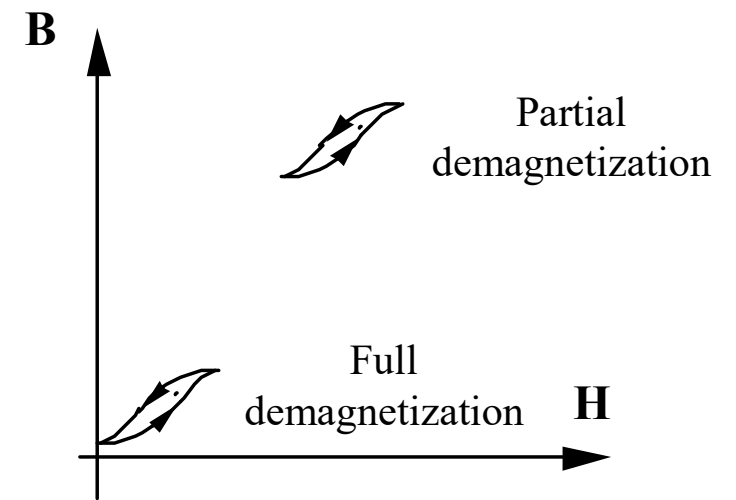
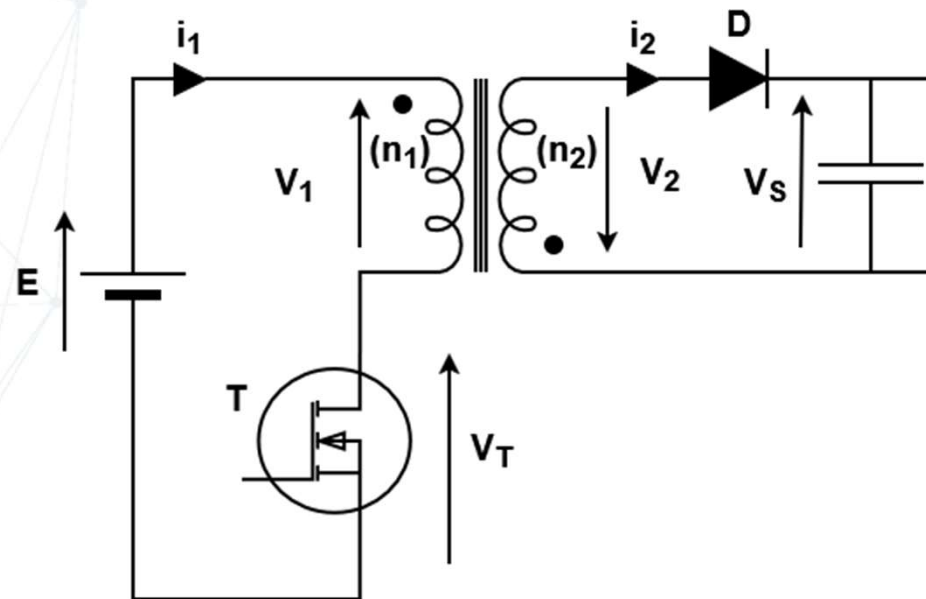
Introduction

Flyback

Forward

Push-Pull

Flyback principle



Easiness, low power, few components

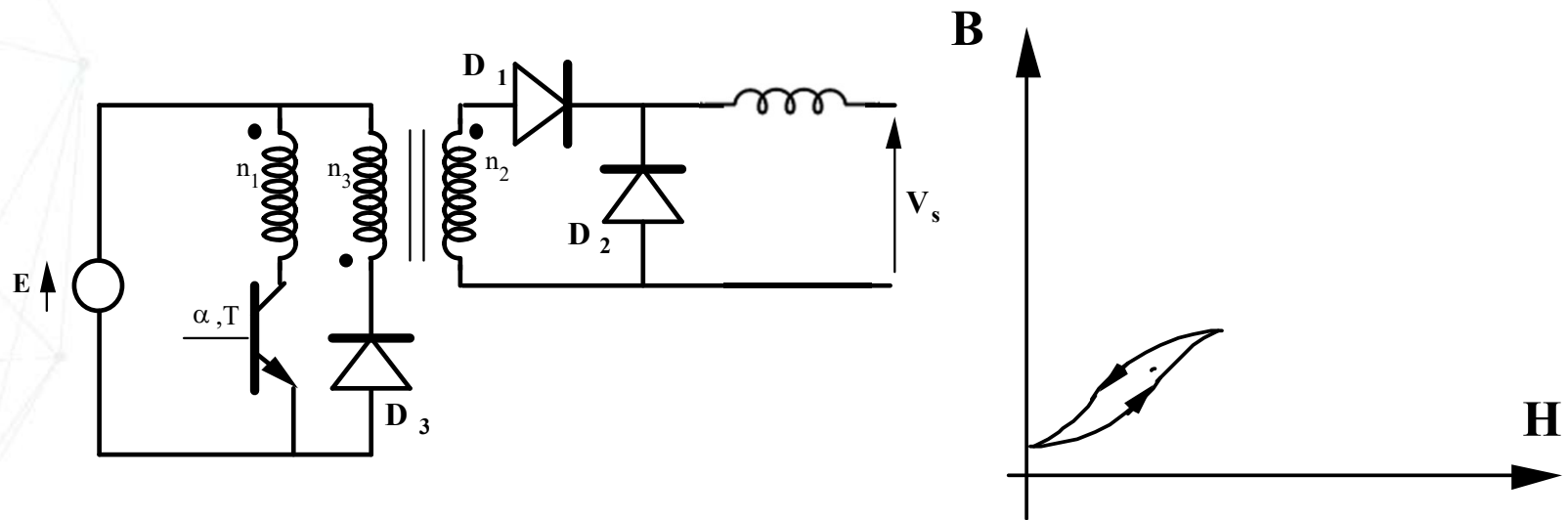
Introduction

Flyback

Forward

Push-Pull

Forward principle



Magnetic component optimized, more components

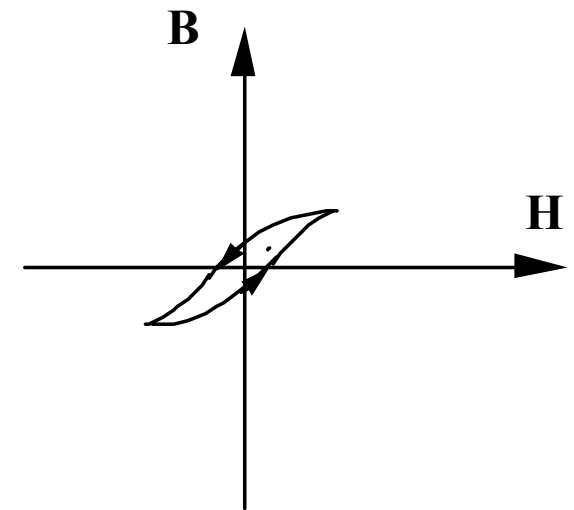
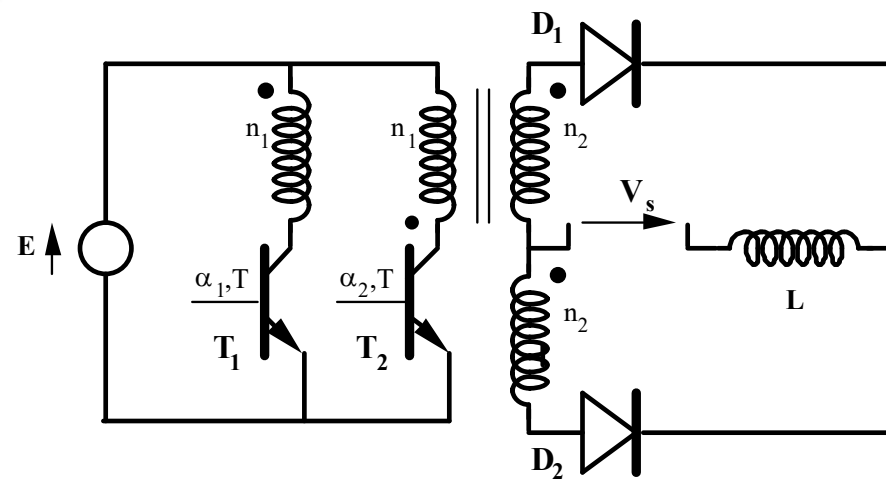
Introduction

Flyback

Forward

Push-Pull

Symmetric converter



High power, 2 transistors or more

Introduction

Flyback

Forward

Push-Pull

Flyback

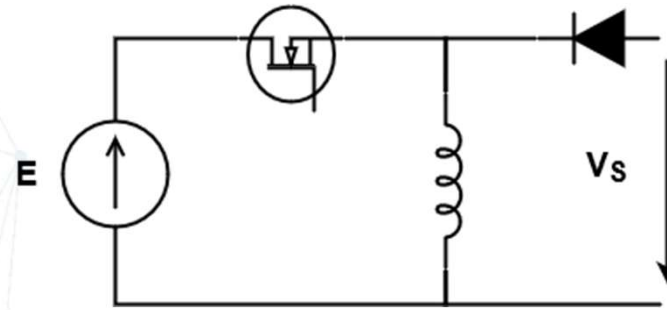
Introduction

Flyback

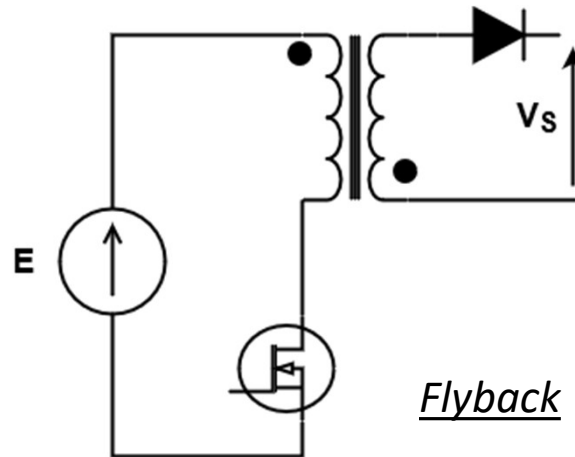
Forward

Push-Pull

FLYBACK converter study

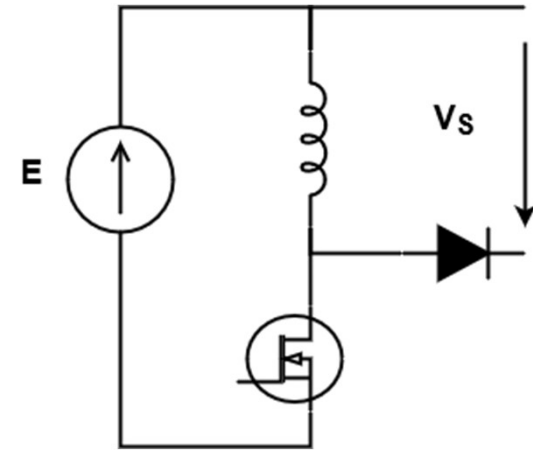


Buck - Boost

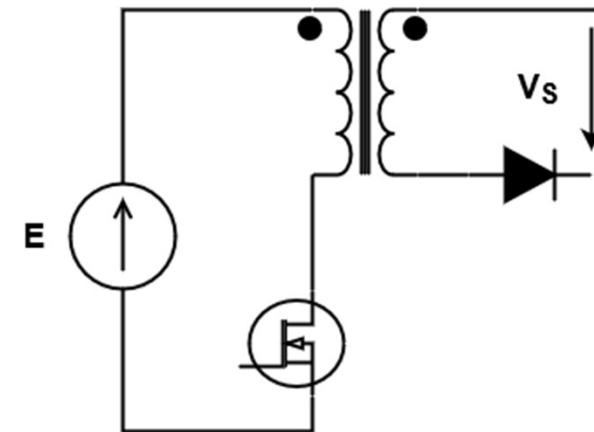


Flyback

≡



≡



Introduction

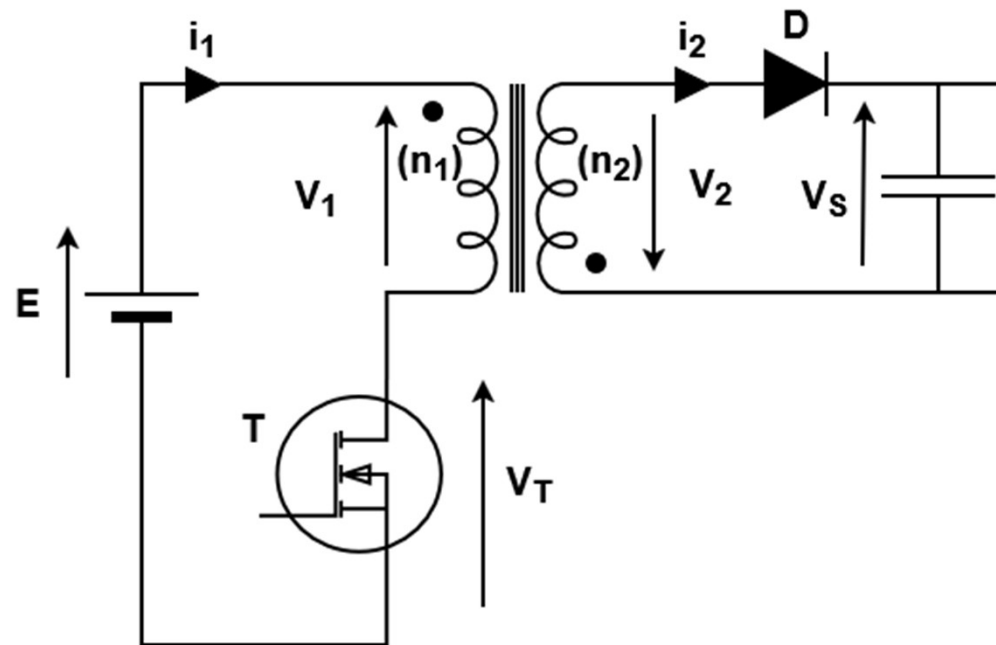
Flyback Transformer

Continuous mode
Discontinuous mode
Sizing

Forward

Push-Pull

FLYBACK converter study



Hypothesis : E constant, V_s constant, perfect semi-conductors

2 magnetic modes

Partial demagnetization (continuous mode)

Full demagnetization (discontinuous mode)

Introduction

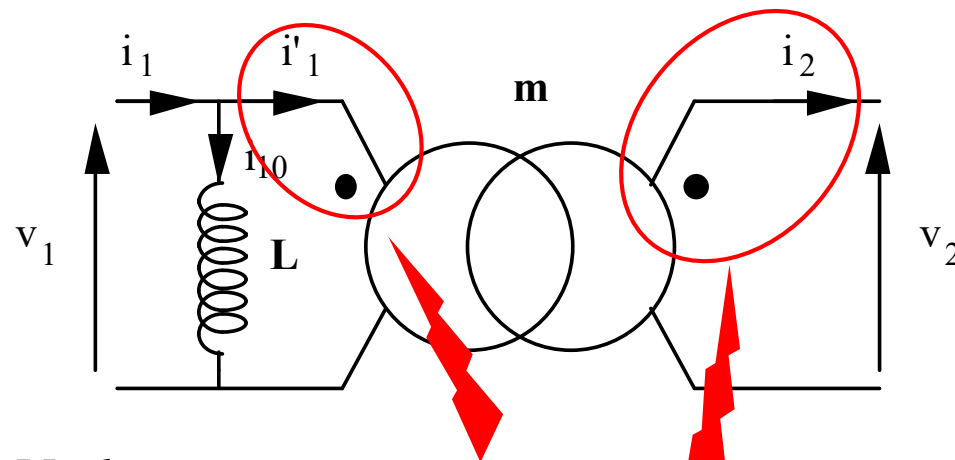
Flyback Transformer

Continuous mode
Discontinuous mode
Sizing

Forward

Push-Pull

Transformer model: convention !



No losses

L : magnetizing inductance (at the primary)

R : Reluctance of the magnetic circuit

$$m = \frac{n_2}{n_1} = \frac{v_2}{v_1} = \frac{i_1'}{i_2}$$

$$\oint H \cdot dl = R \cdot \varphi = n_1 \cdot i_1 - n_2 \cdot i_2 = n_1 \cdot i_{10}$$

Hopkinson's law

Introduction

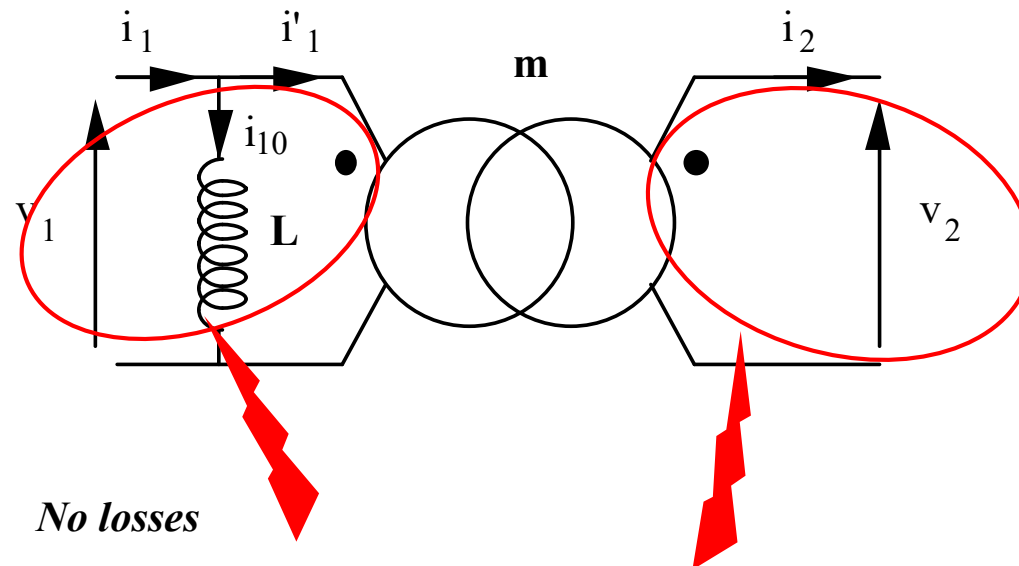
Flyback Transformer

Continuous mode
Discontinuous mode
Sizing

Forward

Push-Pull

Transformer model: convention !



$$v_1 = +n_1 \frac{d\phi}{dt} \quad v_2 = +n_2 \frac{d\phi}{dt}$$

Introduction

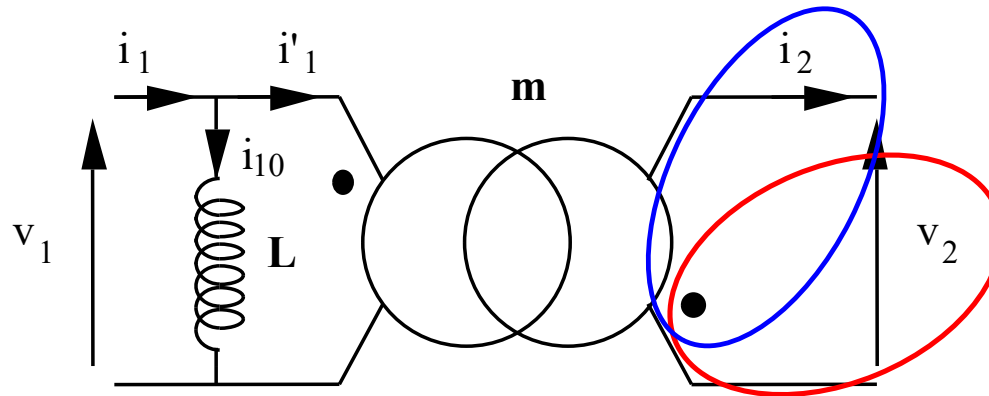
Flyback Transformer

Continuous mode
Discontinuous mode
Sizing

Forward

Push-Pull

FLYBACK transformer model



$$m = \frac{N_2}{N_1} = -\frac{v_2}{v_1} = -\frac{i'_1}{i_2}$$

$$v_2 = -n_2 \frac{d\phi}{dt}$$

$$\oint H \cdot dl = R \cdot \phi = n_1 \cdot i_1 + n_2 \cdot i_2 = n_1 \cdot i_{10}$$

Introduction

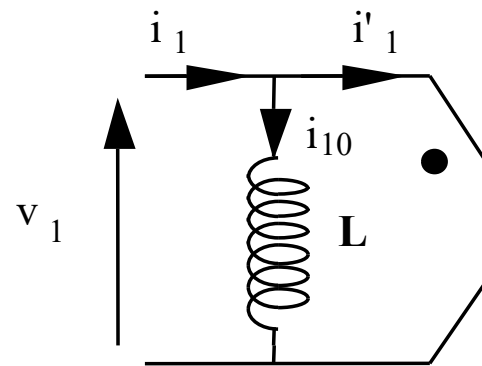
Flyback Transformer

Continuous mode
Discontinuous mode
Sizing

Forward

Push-Pull

Magnetizing current



$$i_1 = i_{10} + i'_1$$

$$R \cdot \varphi = n_1 \cdot i_{10}$$

$$L_1 = \frac{n_1^2}{R}$$

Magnetizing current $\Leftrightarrow$ flux in the magnetic circuit

Introduction

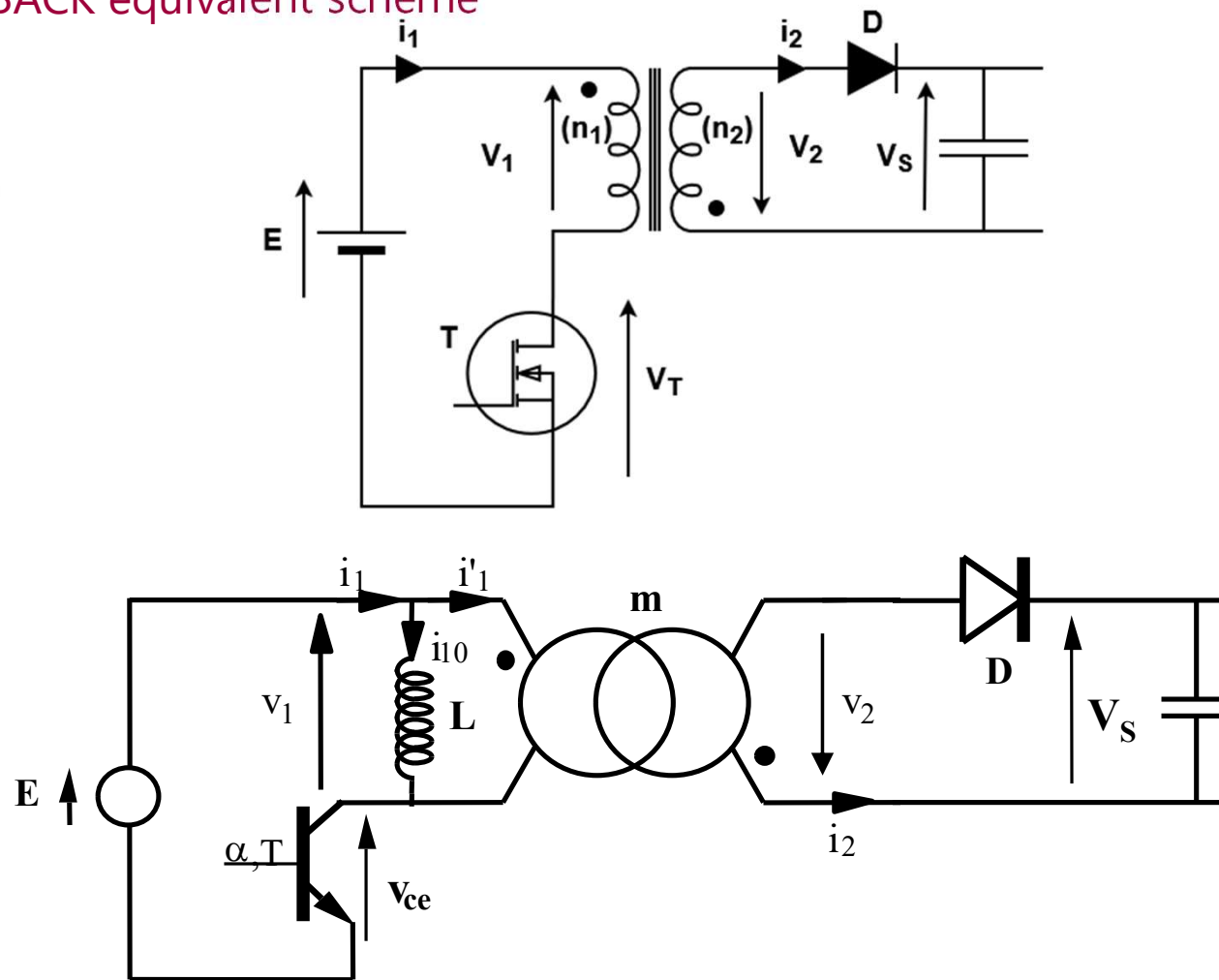
Flyback Transformer

Continuous mode
Discontinuous mode
Sizing

Forward

Push-Pull

FLYBACK equivalent scheme



Introduction

Flyback

Transformer

Continuous mode

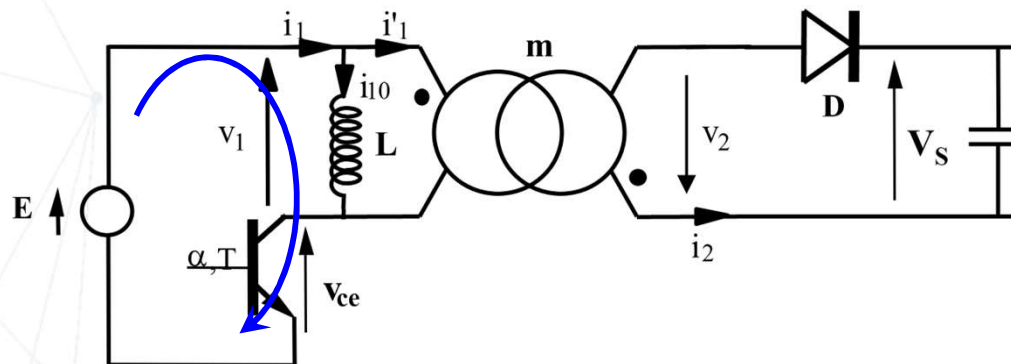
Discontinuous mode

Sizing

Forward

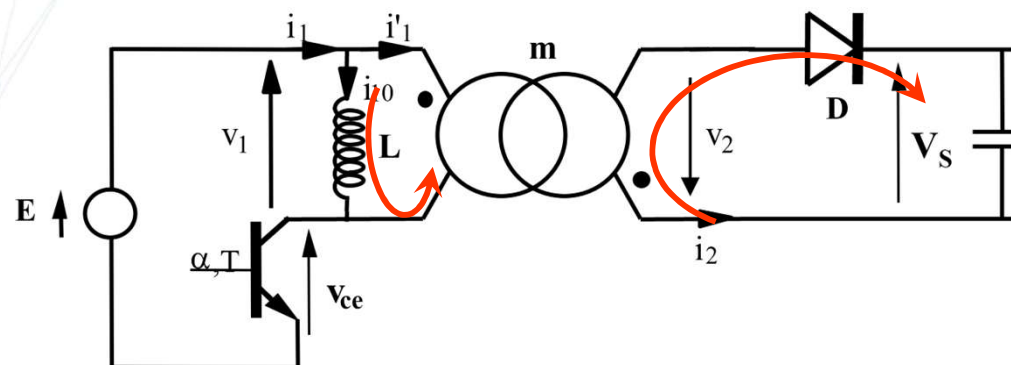
Push-Pull

Partial demagnetization mode: continuous conduction



$0 < t < \alpha T : T_p \text{ closed}$
 $\Rightarrow v_1 = E \Rightarrow v_2 = m.E$
 $v_D = -mE - V_s < 0 \Rightarrow D \text{ reverse}$

$i_2 = 0 \Rightarrow i'_1 = 0 \Rightarrow i_1 = i_{10}$
Energy storing



$\alpha T < t < T : T_p \text{ opened}$
 $\Rightarrow i_1 = 0 \Rightarrow i'_1 = -i_{10} < 0$
 $\Rightarrow i_2 < 0 \Rightarrow D \text{ forward}$

$v_2 = -V_s \Rightarrow v_1 = -V_s / m \Rightarrow \phi \text{ decreases}$
Energy restitution

$$\oint H \cdot dl = R \cdot \phi = n_1 \cdot i_1 - n_2 \cdot i_2 = n_1 \cdot i_{10}$$

Introduction

Flyback

Transformer

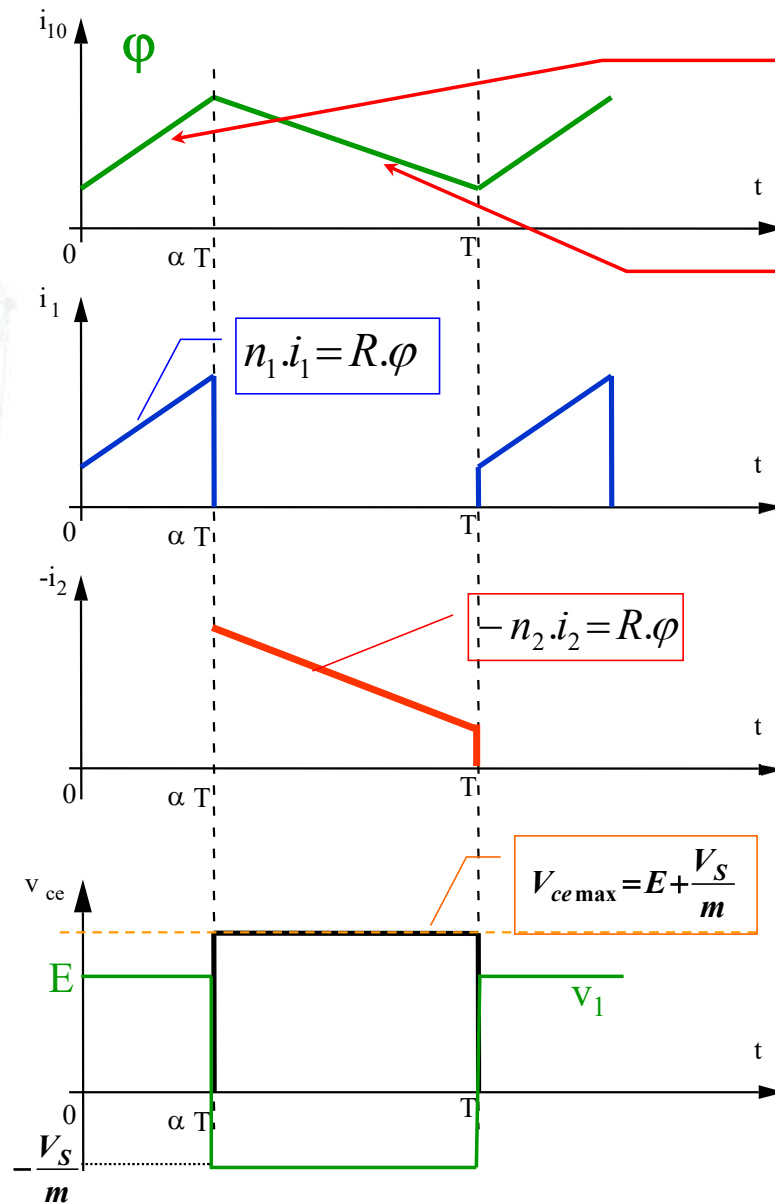
Continuous mode

Discontinuous mode

Sizing

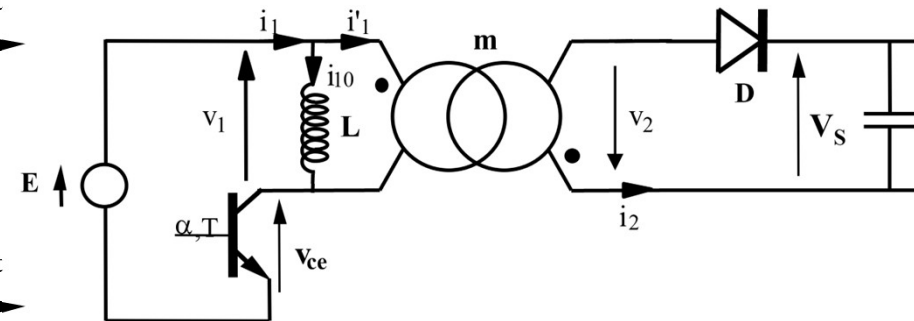
Forward

Push-Pull



$$v_1 = L_1 \cdot \frac{di_{10}}{dt} = n_1 \cdot \frac{d\phi}{dt} \quad \frac{d\phi}{dt} = \frac{E}{n_1}$$

$$v_2 = n_2 \cdot \frac{d\phi}{dt} \quad \frac{d\phi}{dt} = -\frac{V_S}{n_2}$$



Output voltage calculation

$$\langle v_1 \rangle = 0 \quad \langle v_1 \rangle = \frac{1}{T} \cdot \left[E \cdot \alpha T + \left(-\frac{V_S}{m} \right) \cdot (T - \alpha T) \right]$$

$$V_S = \frac{\alpha}{1-\alpha} \cdot mE$$

Introduction

Flyback

Transformer

Continuous mode

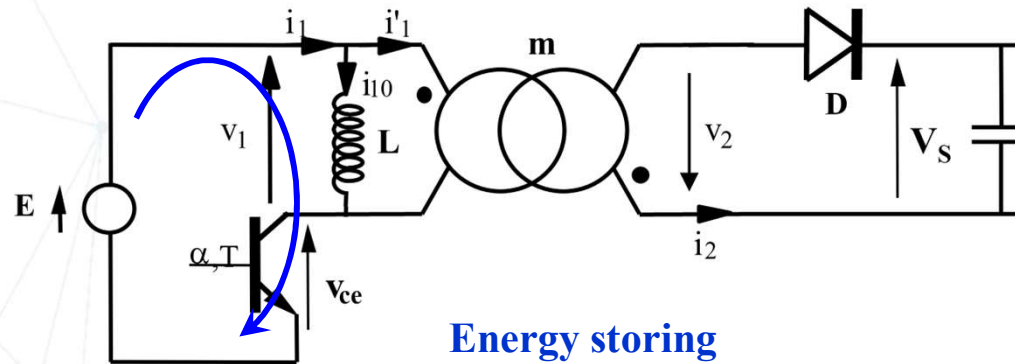
Discontinuous mode

Sizing

Forward

Push-Pull

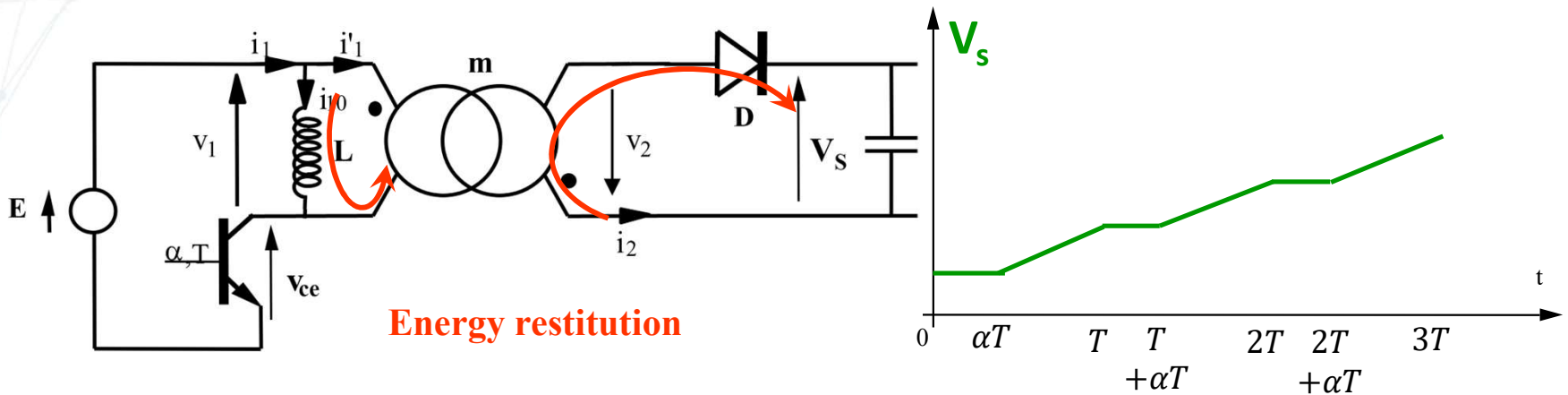
Demagnetization issue



Warning:

If no load connected
-> Output capacitor never
discharges

-> Issue



Introduction

Flyback

Transformer

Continuous mode

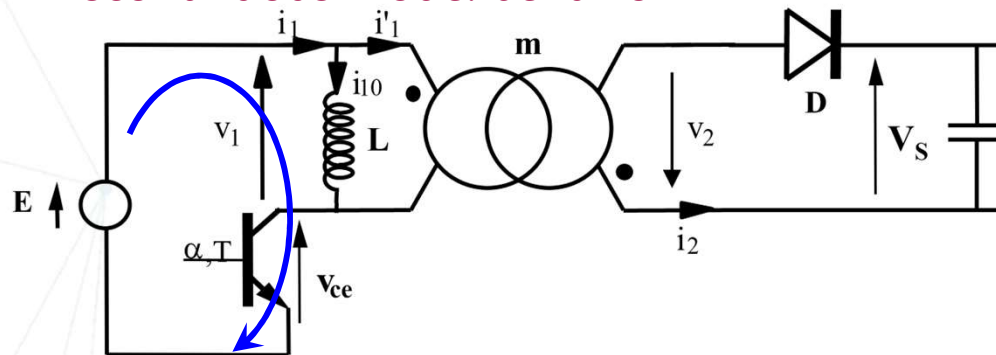
Discontinuous mode

Sizing

Forward

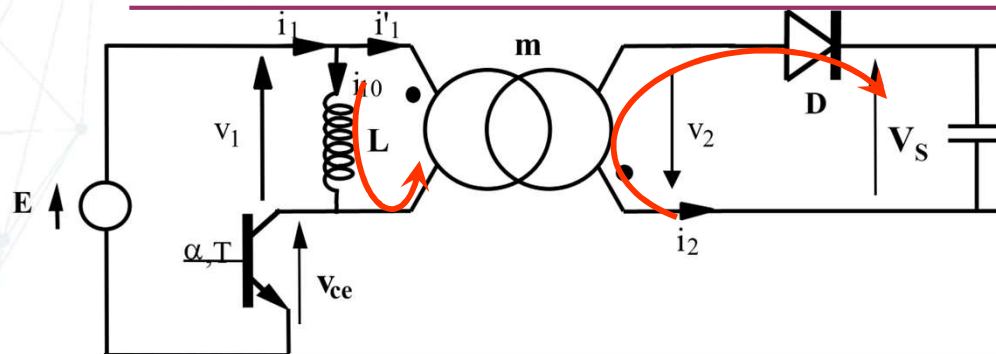
Push-Pull

Discontinuous mode: behavior



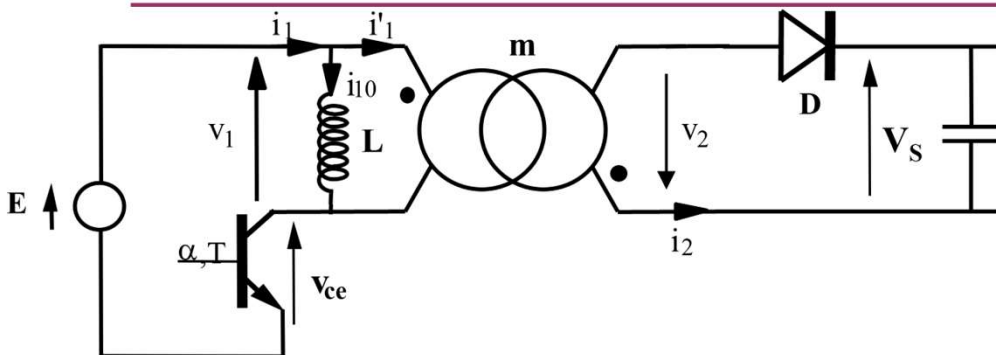
$0 < t < \alpha T : T_p \text{ closed}$
 $\Rightarrow v_1 = E \Rightarrow v_2 = m.E$
 $v_D < 0 \Rightarrow D \text{ reverse}$

$i_2 = 0 \Rightarrow i'_1 = 0 \Rightarrow i_1 = i_{10}$
Energy storage
Unchanged mode



$\alpha T < t < \alpha' T : T_p \text{ opened}$
 $\Rightarrow i_1 = 0 \Rightarrow i'_1 = -i_{10} < 0$
 $\Rightarrow i_2 < 0 \Rightarrow D \text{ forward}$
and then reverse naturally

$v_2 = -V_S \Rightarrow v_1 = -V_S/m \Rightarrow \phi \text{ decreases}$
Energy restitution



$\alpha' T < t < T : T \text{ and } D \text{ both reverse}$
All currents are zero
No more energy
 $\Rightarrow v_1 = 0 ; v_2 = 0 \Rightarrow \phi = 0$

Introduction

Flyback

Transformer

Continuous mode

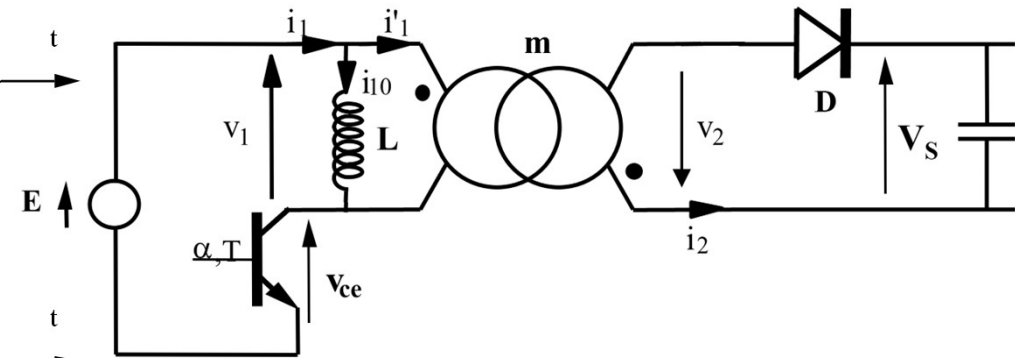
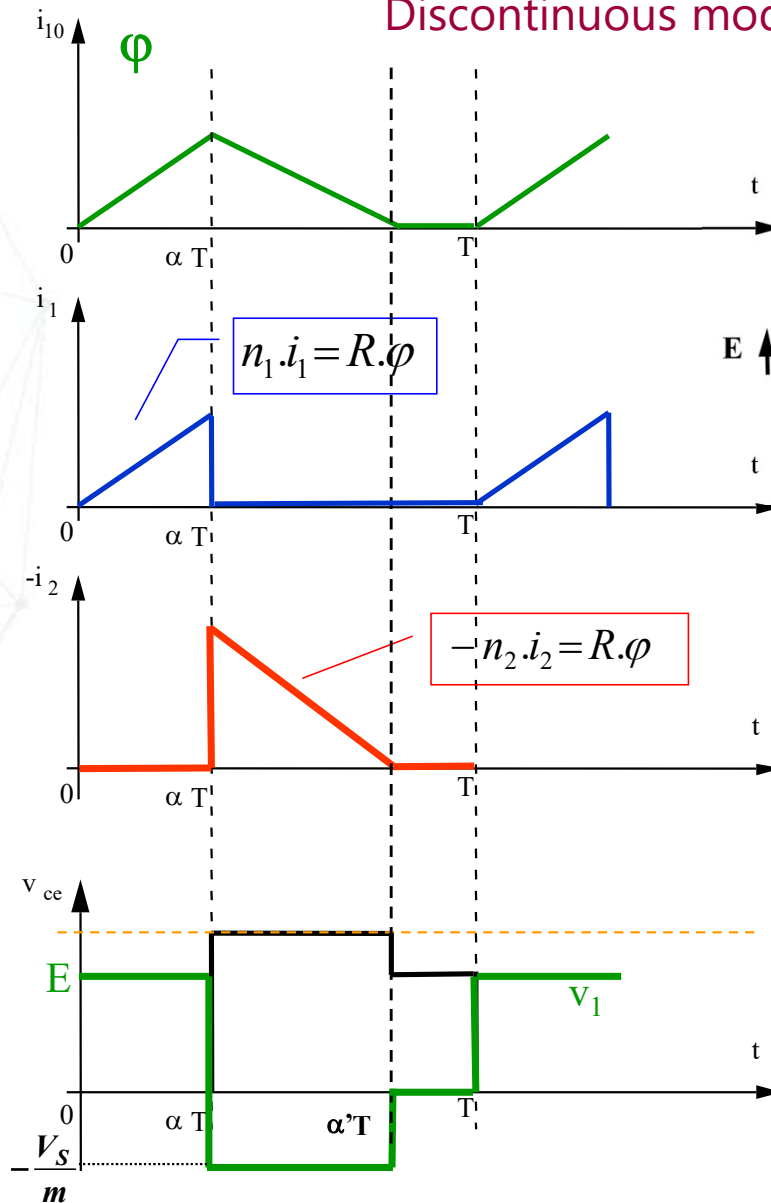
Discontinuous mode

Sizing

Forward

Push-Pull

Discontinuous mode: chronograms



Output voltage calculation:
Transferred energy

$$W_L = \frac{1}{2} \cdot L \cdot I_{1\max}^2 = \frac{1}{2} \cdot L \cdot \left(\frac{\alpha \cdot E \cdot T}{L} \right)^2$$

$$P_S = V_S \cdot I_S = f \cdot W_L$$

$$V_S = \frac{\alpha^2 \cdot E^2 \cdot T}{2 \cdot L \cdot I_S}$$

Introduction

Flyback

Transformer

Continuous mode

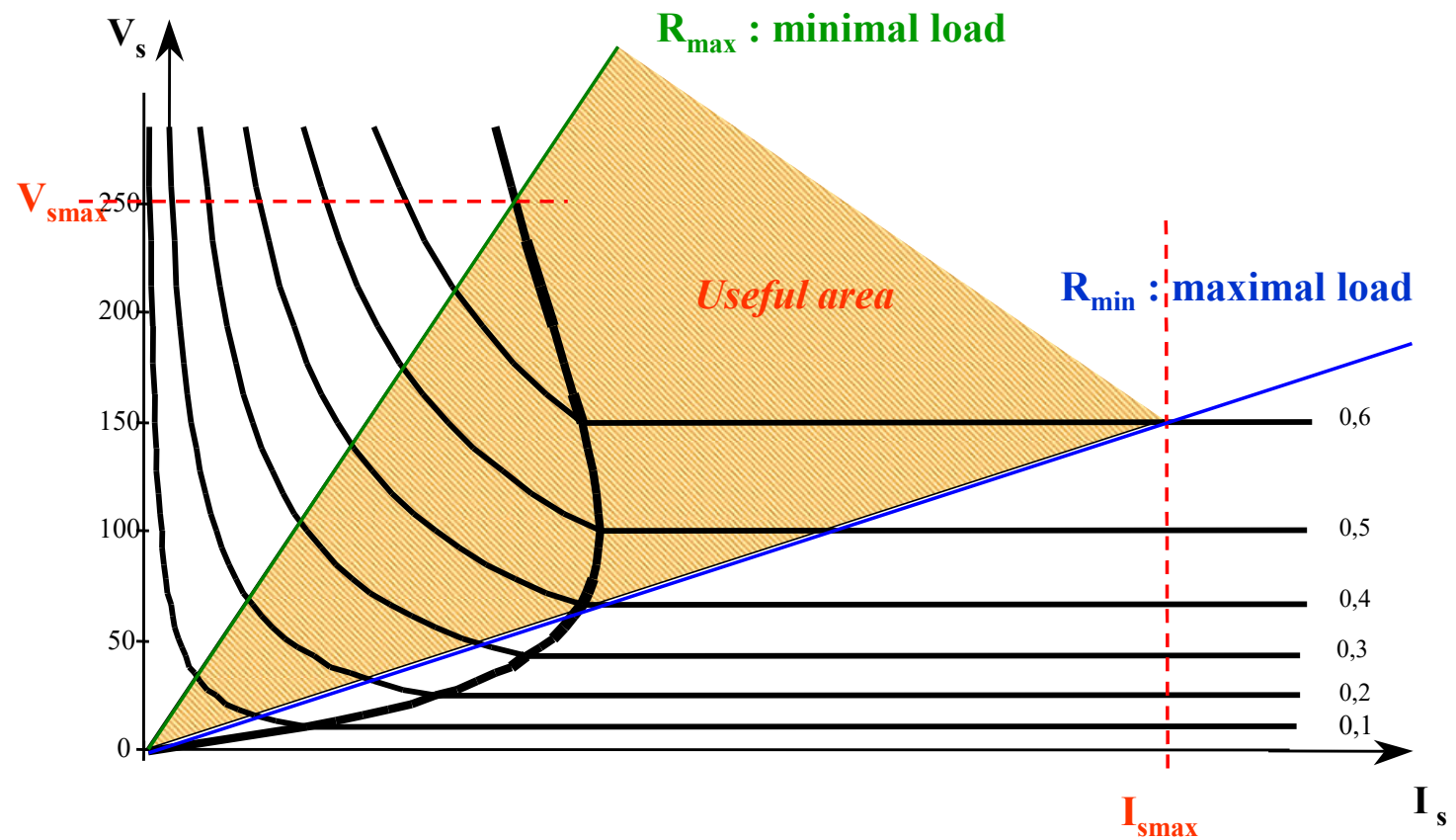
Discontinuous mode

Sizing

Forward

Push-Pull

Discontinuous mode: output characteristics



Introduction

Flyback

Transformer

Continuous mode

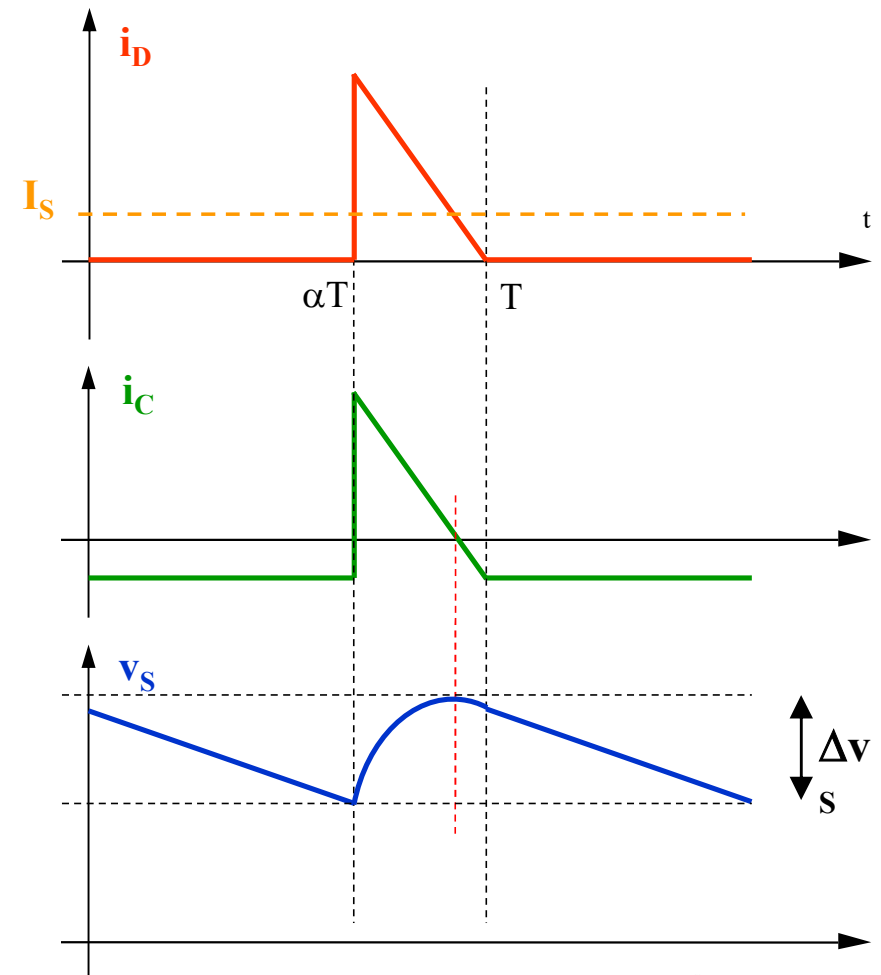
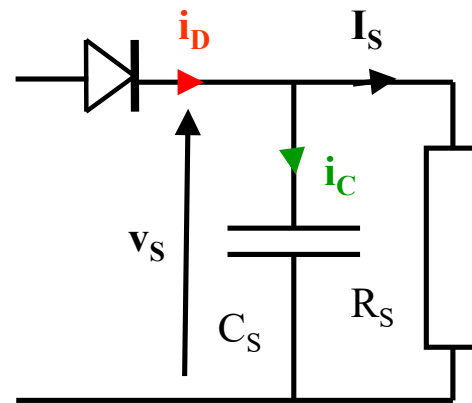
Discontinuous mode

Sizing

Forward

Push-Pull

Boundary mode: Output filtering



Introduction

Flyback

Transformer

Continuous mode

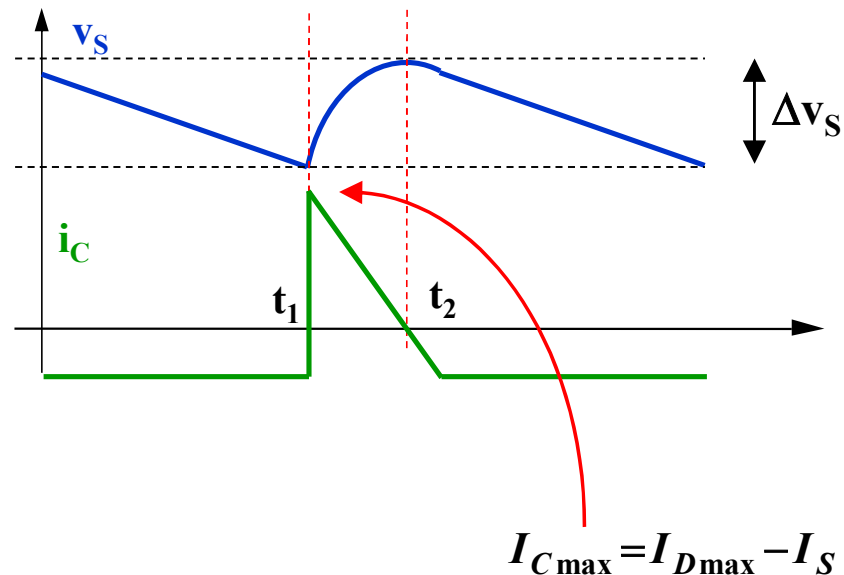
Discontinuous mode

Sizing

Forward

Push-Pull

Boundary mode: capacitor calculation

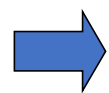


$$\Delta v_s = \frac{1}{C_s} \int_{t_1}^{t_2} i_c(t) dt$$

$$t_1 = \alpha T$$

$$I_{Dmax} = \frac{n_1}{n_2} \frac{\alpha \cdot ET}{L}$$

$$I_{Cmax} = I_{Dmax} - I_S \quad i_c(t_2) = 0$$



$$C_s = \frac{\alpha^2 \cdot E \cdot T}{(1 - \alpha) \cdot R_s \cdot \Delta V_s} \quad \text{same as Buck-Boost}$$

For high value of C_s ,
 $\Delta v_s \approx \Delta v_{ESR} = ESR \cdot \Delta i_c = ESR \cdot \Delta i_L$

$$ESR < \frac{\Delta V_s}{\Delta i_L}$$

Introduction

Flyback

Transformer

Continuous mode

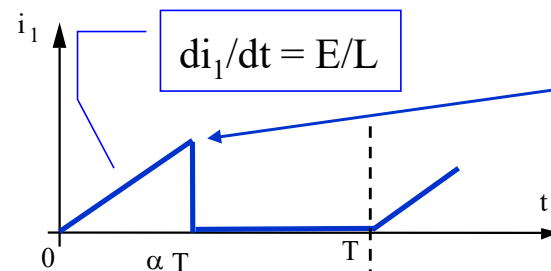
Discontinuous mode

Sizing

Forward

Push-Pull

Boundary mode: sizing components



$$I_{1\max} = \frac{\alpha \cdot E \cdot T}{L}$$

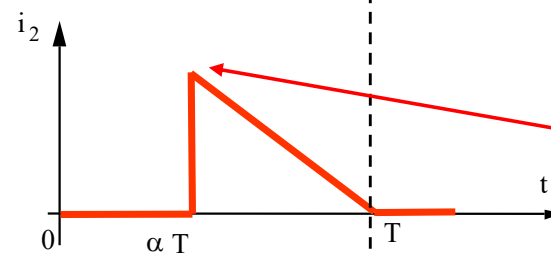
$$I_{1\text{moy}} = \frac{\alpha^2 \cdot E \cdot T}{2L}$$

$$I_{1\text{eff}}^2 = \frac{1}{T} \cdot \int_0^T i_1^2(t) \cdot dt = \frac{1}{T} \cdot \left(\frac{E}{L} \right)^2 \cdot \frac{\alpha^3 T^3}{3} = \frac{\alpha}{3} \cdot I_{1\max}^2$$

$$I_{1\text{eff}} = \sqrt{\frac{\alpha}{3}} \cdot I_{1\max}$$

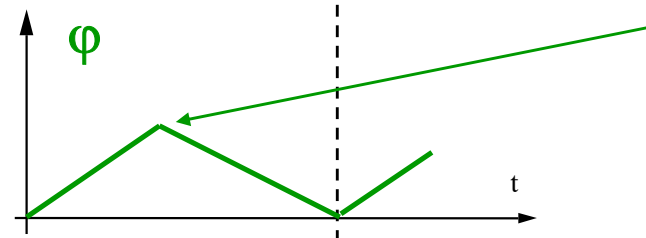
$$I_{2\max} = \frac{n_1}{n_2} \cdot I_{1\max}$$

$$I_{2\text{eff}} = \sqrt{\frac{1-\alpha}{3}} \cdot I_{2\max}$$



$$\varphi_{\max} = \frac{n_1 I_{1\max}}{R}$$

Non-saturation : $\varphi_{\max} = A_e B_{\text{sat}}$



$$I_{1\max} < \frac{R \cdot B_{\text{sat}} \cdot A_e}{n_1}$$

Introduction

Flyback

Transformer

Continuous mode

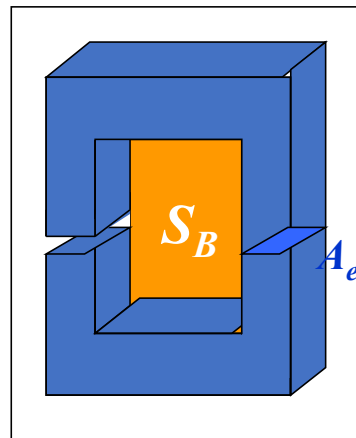
Discontinuous mode

Sizing

Forward

Push-Pull

Boundary mode: transformer sizing



$$I_{1eff} = \sqrt{\frac{\alpha}{3}} \cdot I_{1max}$$

$$I_{2eff} = \sqrt{\frac{1-\alpha}{3}} \cdot I_{2max}$$

$$I_{2max} = \frac{n_1}{n_2} \cdot I_{1max}$$



$$\delta = \frac{I_{1eff}}{S_{sp1}} = \frac{I_{2eff}}{S_{sp2}} < \text{max current density: } \delta_{max}$$

$$S_{Tcu} = n_1 \cdot S_{sp1} + n_2 \cdot S_{sp2} < k_B \cdot \text{Window area } S_B$$

k_B = winding coefficient

$$k_B \cdot S_B > n_1 \cdot \frac{I_{1eff}}{\delta_{max}} + n_2 \cdot \frac{I_{2eff}}{\delta_{max}}$$

$$I_{1max} < \frac{k_B \cdot S_B \cdot \delta_{max}}{n_1} \cdot \frac{\sqrt{3}}{\sqrt{\alpha} + \sqrt{1-\alpha}}$$

Introduction

Flyback

Transformer

Continuous mode

Discontinuous mode

Sizing

Forward

Push-Pull

Boundary mode: transformer sizing

$$P_S = f \cdot W_L = f \cdot \frac{1}{2} \cdot L \cdot I_{1\max}^2$$
$$I_{1\max} < \frac{R \cdot B_{sat} \cdot A_e}{n_1}$$
$$I_{1\max} < \frac{k_B \cdot S_B \cdot \delta_{\max}}{n_1} \cdot \frac{\sqrt{3}}{\sqrt{\alpha} + \sqrt{1-\alpha}}$$
$$P_S < f \cdot \frac{L}{2} \cdot \frac{R \cdot B_{sat} \cdot A_e}{n_1} \cdot \frac{k_B \cdot S_B \cdot \delta_{\max}}{n_1} \cdot \frac{\sqrt{3}}{\sqrt{\alpha} + \sqrt{1-\alpha}}$$

$$A_e \cdot S_B > 2 \cdot \frac{\sqrt{\alpha} + \sqrt{1-\alpha}}{\sqrt{3}} \cdot \frac{P_S}{f \cdot k_B \cdot B_{sat} \cdot \delta_{\max}}$$

 *With P_S*

 *With f*

Introduction

Flyback

Transformer

Continuous mode

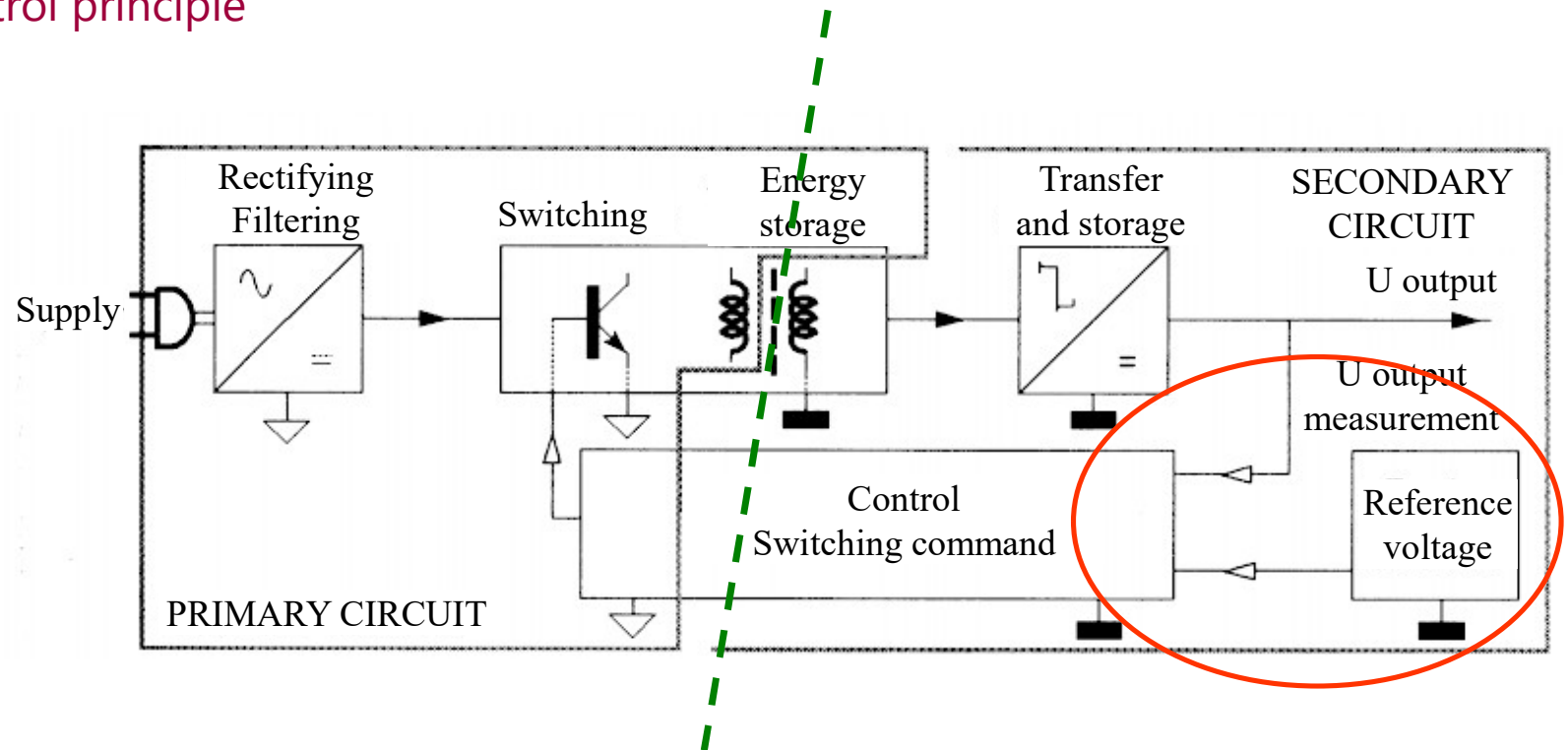
Discontinuous mode

Sizing

Forward

Push-Pull

Control principle



Output voltage control

Introduction

Flyback

Transformer

Continuous mode

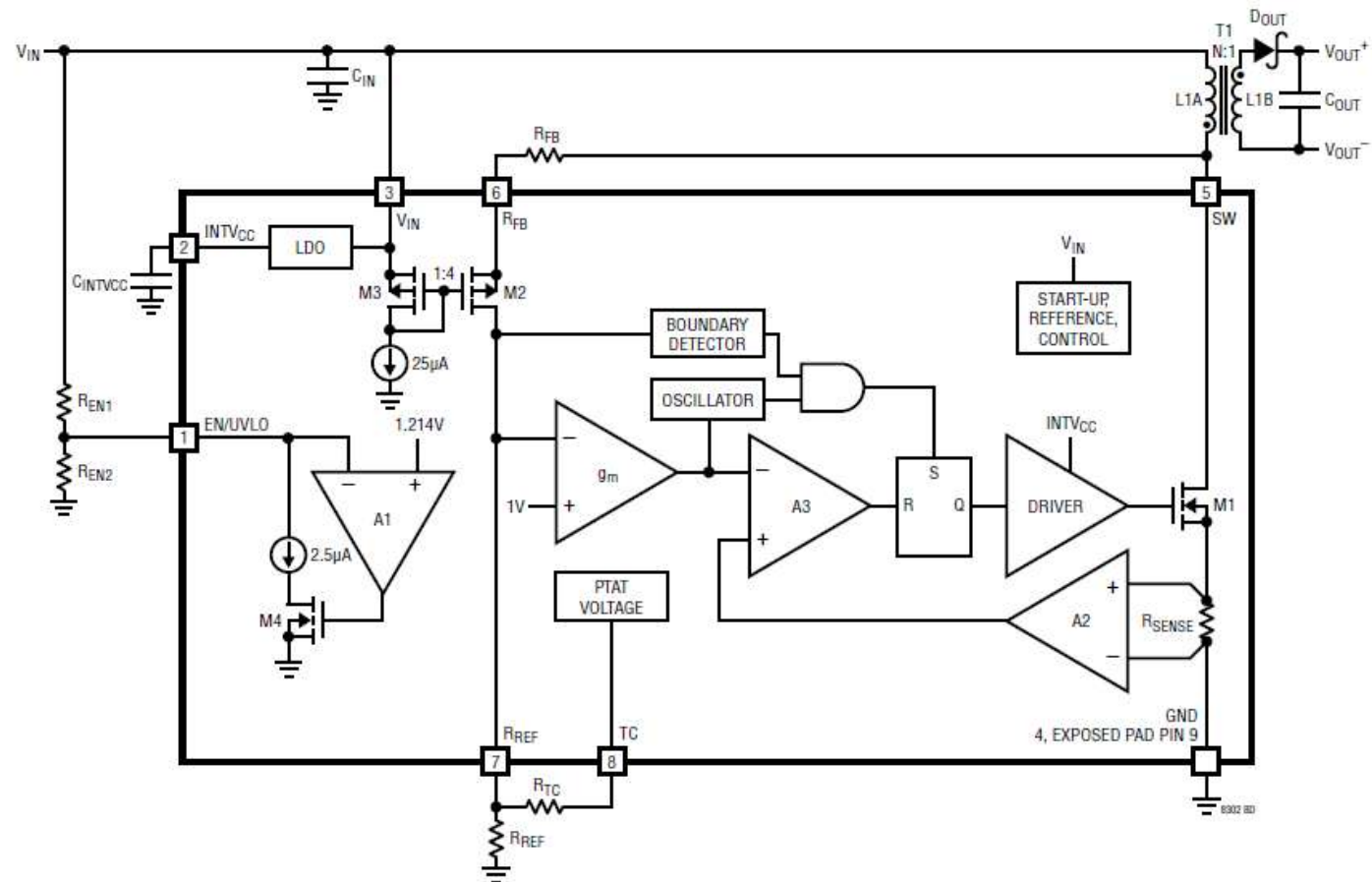
Discontinuous mode

Sizing

Forward

Push-Pull

Example: LT8302 (internal scheme)



Introduction

Flyback

Transformer

Continuous mode

Discontinuous mode

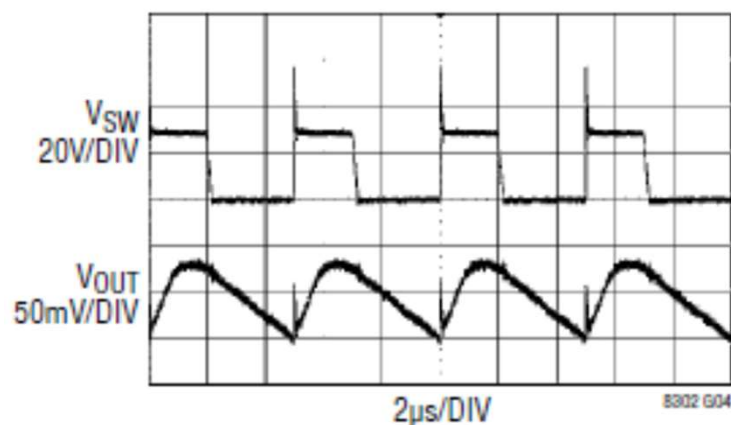
Sizing

Forward

Push-Pull

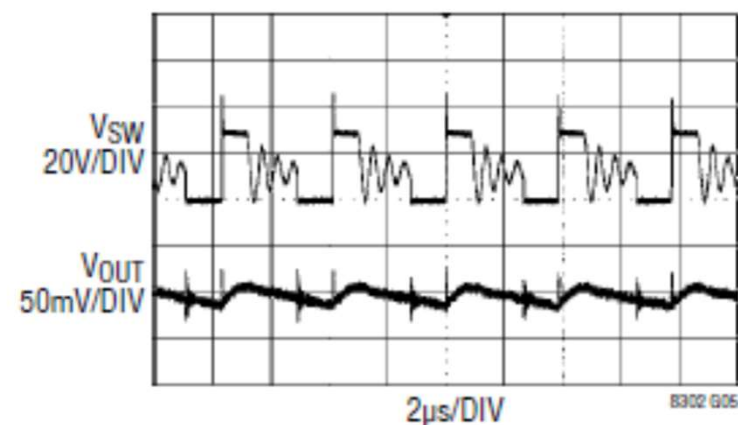
Example: LT8302

Boundary Mode Waveforms



FRONT PAGE APPLICATION
 $V_{IN} = 12V$
 $I_{OUT} = 2A$

Discontinuous Mode Waveforms



FRONT PAGE APPLICATION
 $V_{IN} = 12V$
 $I_{OUT} = 0.5A$

Introduction

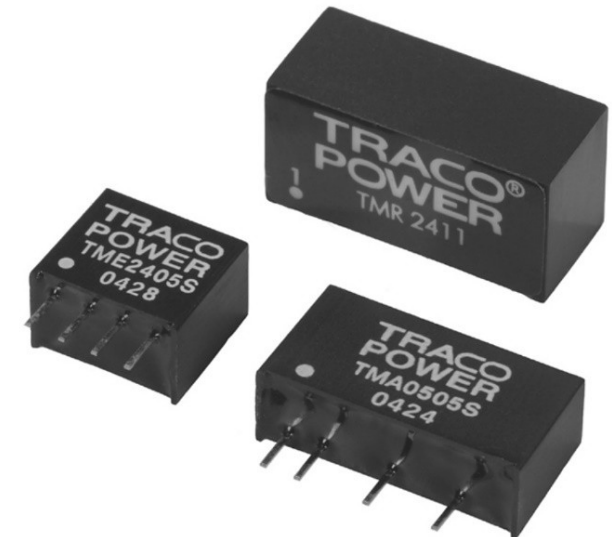
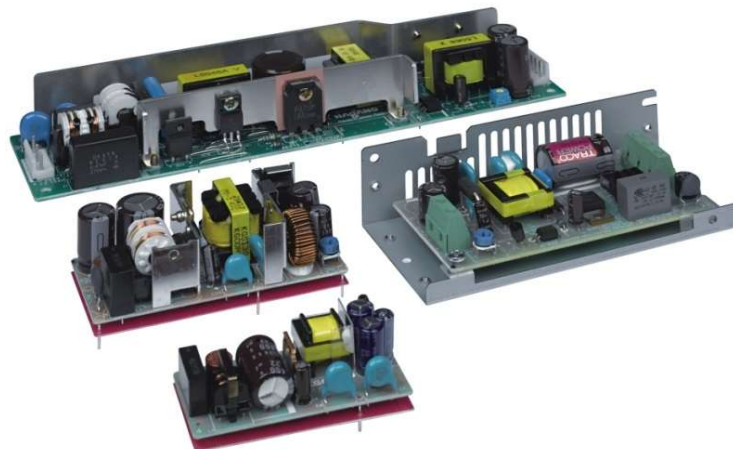
Flyback

Transformer
Continuous mode
Discontinuous mode
Sizing

Forward

Push-Pull

Flyback examples



Introduction

Flyback

Forward

Push-Pull

Forward

Introduction

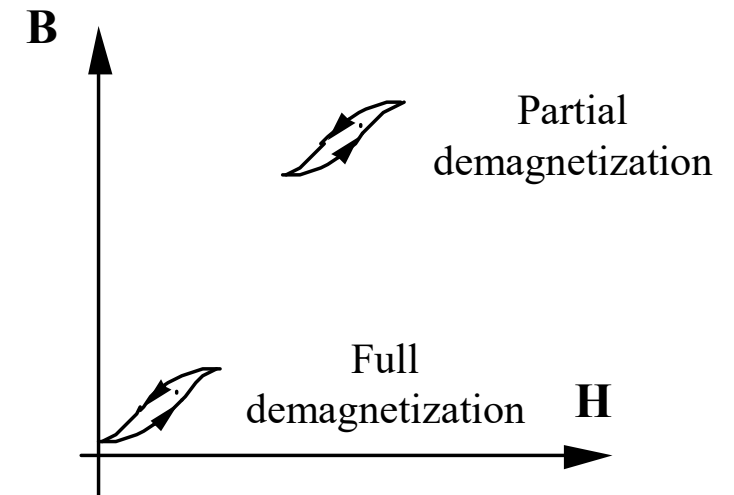
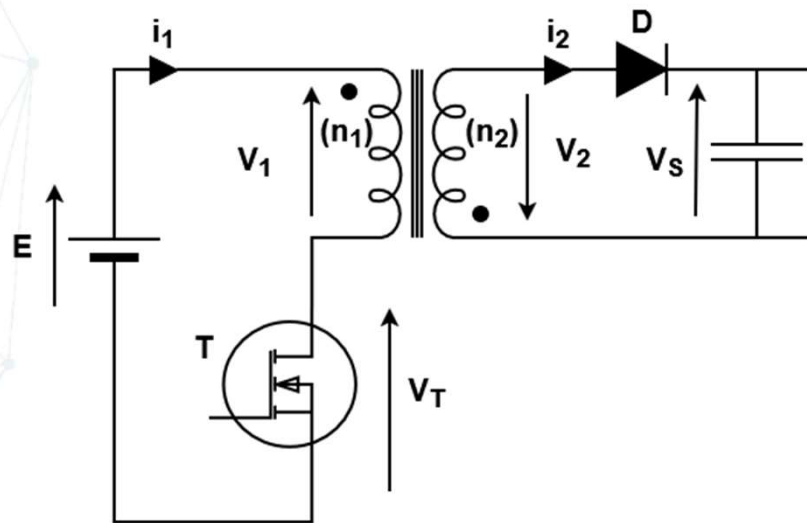
Flyback

**Forward
Principle**

Continuous mode
Sizing

Push-Pull

Flyback principle



Easiness, low power, few components

Limited for high power due to energy storage limitations

Introduction

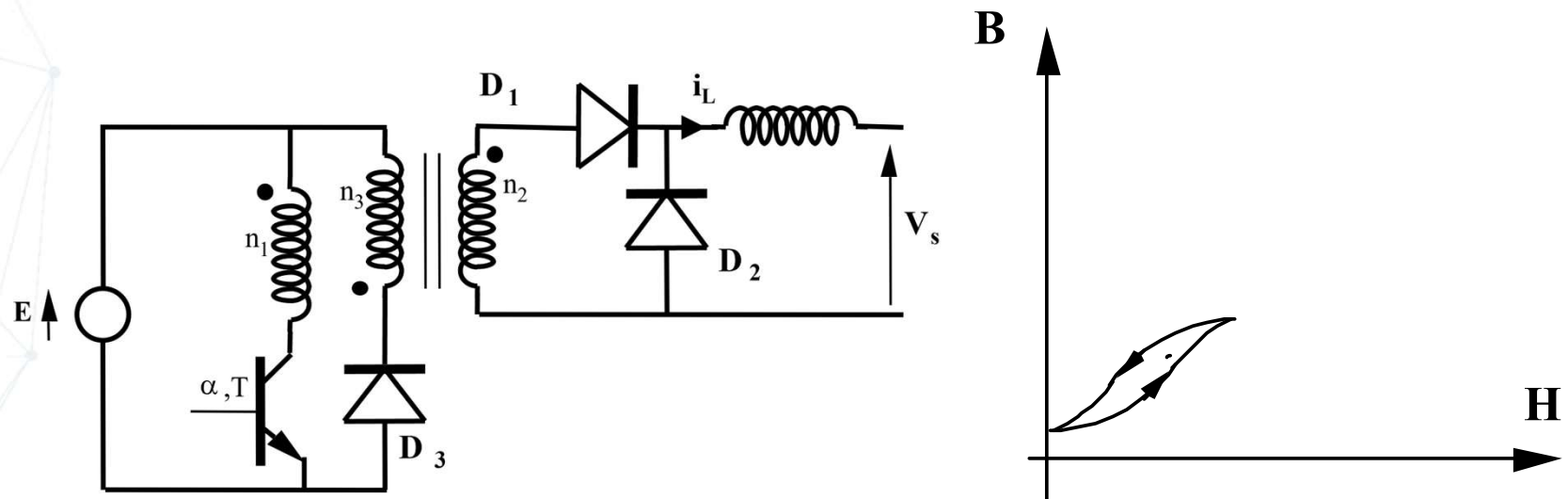
Flyback

**Forward
Principle**

Continuous mode
Sizing

Push-Pull

Forward principle



Magnetic component optimized, more components

Introduction

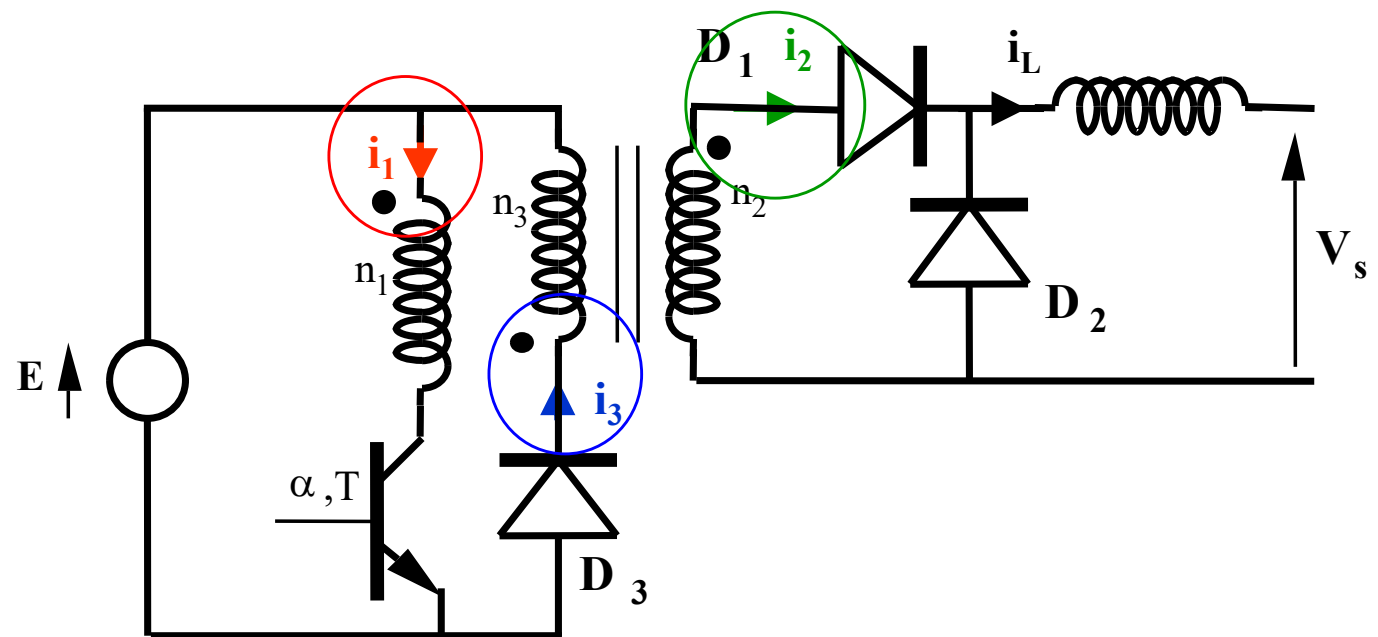
Flyback

Forward
Principle

Continuous mode
Sizing

Push-Pull

Hopkinson's law

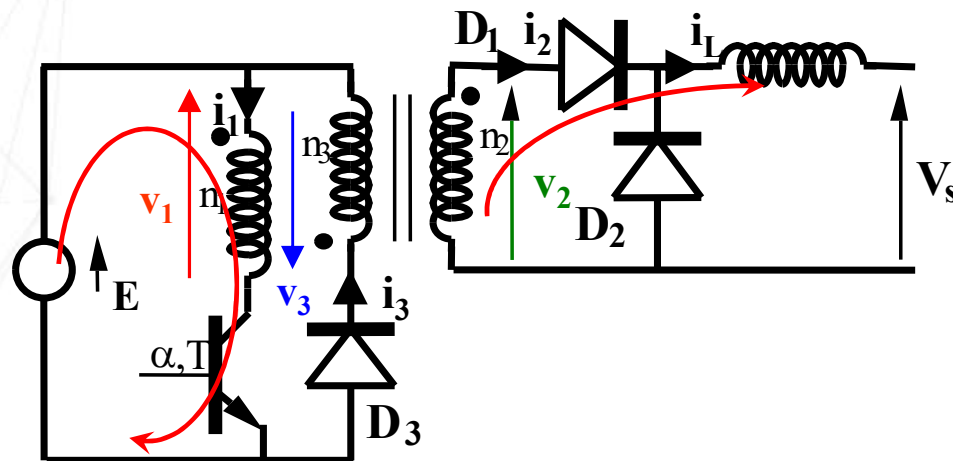


$$\oint H \cdot dl = R \cdot \phi = n_1 \cdot i_1 + n_3 \cdot i_3 - n_2 \cdot i_2$$

Continuous mode study: $i_L > 0$



At least one of the two diodes D_1 and D_2 forwards.



$$v_1 = E = n_1 \cdot \frac{d\phi}{dt}$$

magnetization

$$v_2 = n_2 \cdot \frac{d\phi}{dt}$$

$$v_3 = n_3 \cdot \frac{d\phi}{dt}$$

$$0 < t < \alpha T$$

T_p forward.

$$v_1 = E \Rightarrow v_2 = \frac{n_2}{n_1} \cdot E$$

D_1 forward, D_2 reverse
 $i_2 = i_L > 0$

$$v_1 = E \Rightarrow v_3 = \frac{n_3}{n_1} \cdot E$$

D_3 reverse

$$n_1 \cdot i_1 - n_2 \cdot i_2 = R \cdot \phi$$

Introduction

Flyback

Forward

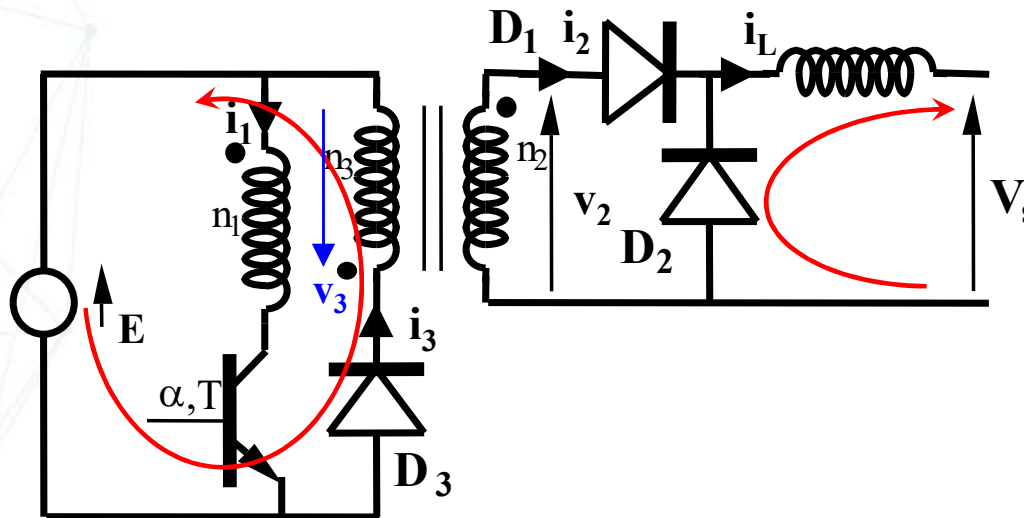
Principle

Continuous mode

Sizing

Push-Pull

Continuous mode study



$$T_p \text{ opened} \Rightarrow i_1 = 0$$

$$n_3 \cdot i_3 - n_2 \cdot i_2 = R \cdot \varphi > 0$$

$$\Rightarrow i_3 > 0 \Rightarrow D_3 \text{ forward}$$

$$v_3 = -E = n_3 \cdot \frac{d\varphi}{dt} \quad \text{demagnetization}$$

$$v_2 = -\frac{n_2}{n_3} E \quad D_1 \text{ reverse, } D_2 \text{ forward} \Rightarrow i_2 = 0$$

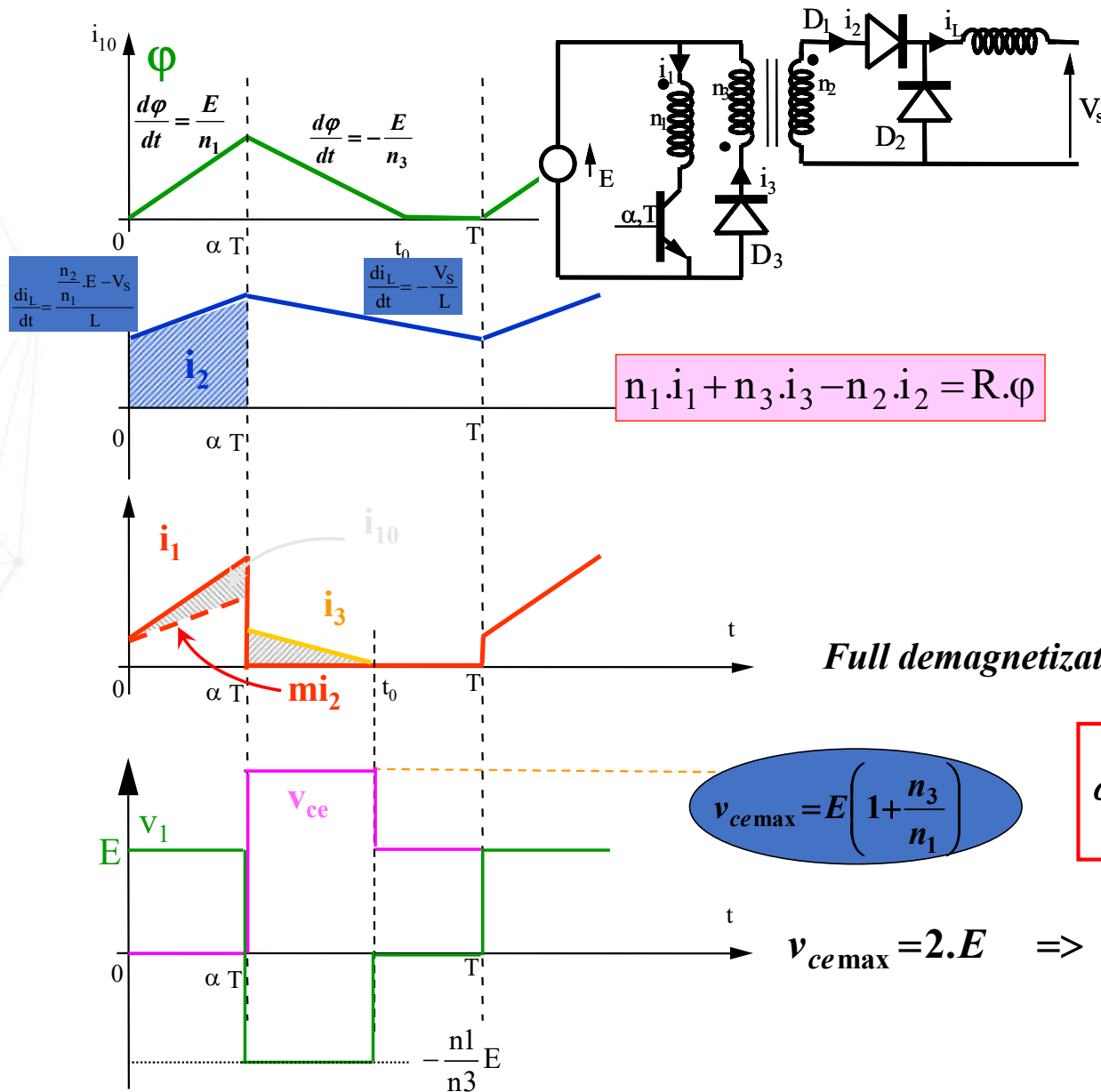
$$L \cdot \frac{di_L}{dt} = -V_s$$

Introduction

Flyback

Forward
Principle
Continuous mode
Sizing

Push-Pull



Introduction

Flyback

Forward

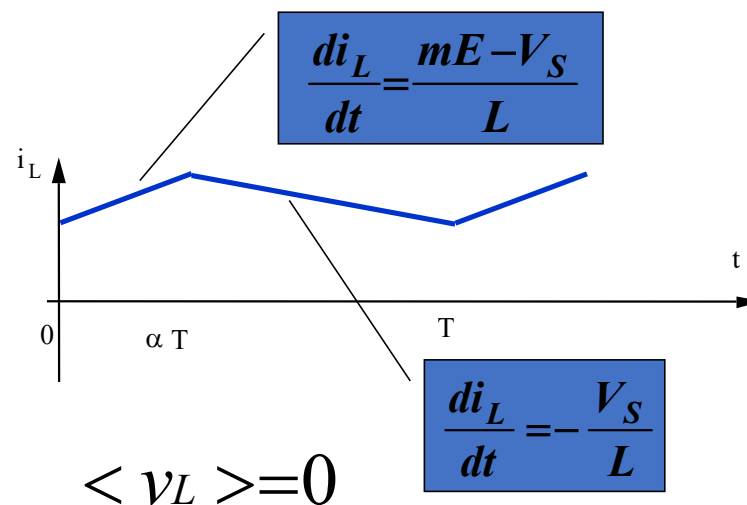
Principle

Continuous mode

Sizing

Push-Pull

Continuous mode study: voltage transfer



$$\Delta i_L = \frac{mE - V_S}{L} \cdot \alpha T = \frac{V_S}{L} (1 - \alpha) T$$

$$V_S = \alpha \cdot m \cdot E$$

Introduction

Flyback

Forward

Principle

Continuous mode

Sizing

Push-Pull

Continuous mode study: Components sizing

- *Filtering capacitor: C_s*
- *Transformer $\Rightarrow Ae.SB$*
- *Output inductance $\Rightarrow Ae.SB$*

Introduction

Flyback

Forward

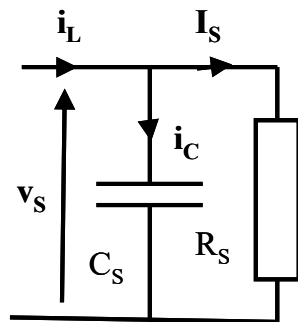
Principle

Continuous mode

Sizing

Push-Pull

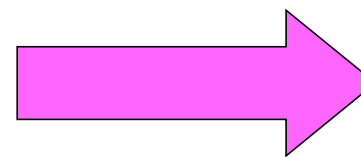
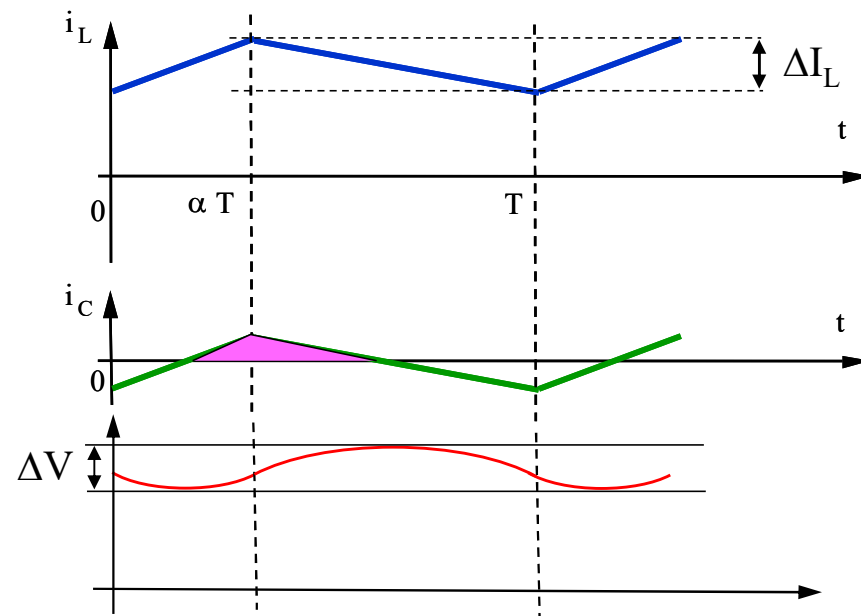
Continuous mode study: Filtering capacitor



$$\Delta V = \frac{1}{C_S} \cdot \int_{\frac{\alpha T}{2}}^{\frac{T+\alpha T}{2}} i_C(t) dt = \frac{1}{C_S} \cdot \frac{1}{2} \cdot i_{C_{\max}} \cdot T$$

$$i_{C_{\max}} = \frac{\Delta I_L}{2} = \frac{1}{2} \cdot \frac{V_S}{L} (1 - \alpha) \cdot T$$

Max ripple for $\alpha = 0.5$



$$C_S = \frac{m \cdot E \cdot T^2}{32 \cdot L \cdot \Delta V}$$

For high value of C_S ,

$$\Delta v_S \approx \Delta v_{ESR} = ESR \cdot \Delta i_C = ESR \cdot \Delta i_L$$

$$ESR < \frac{\Delta V_S}{\Delta i_L}$$

Introduction

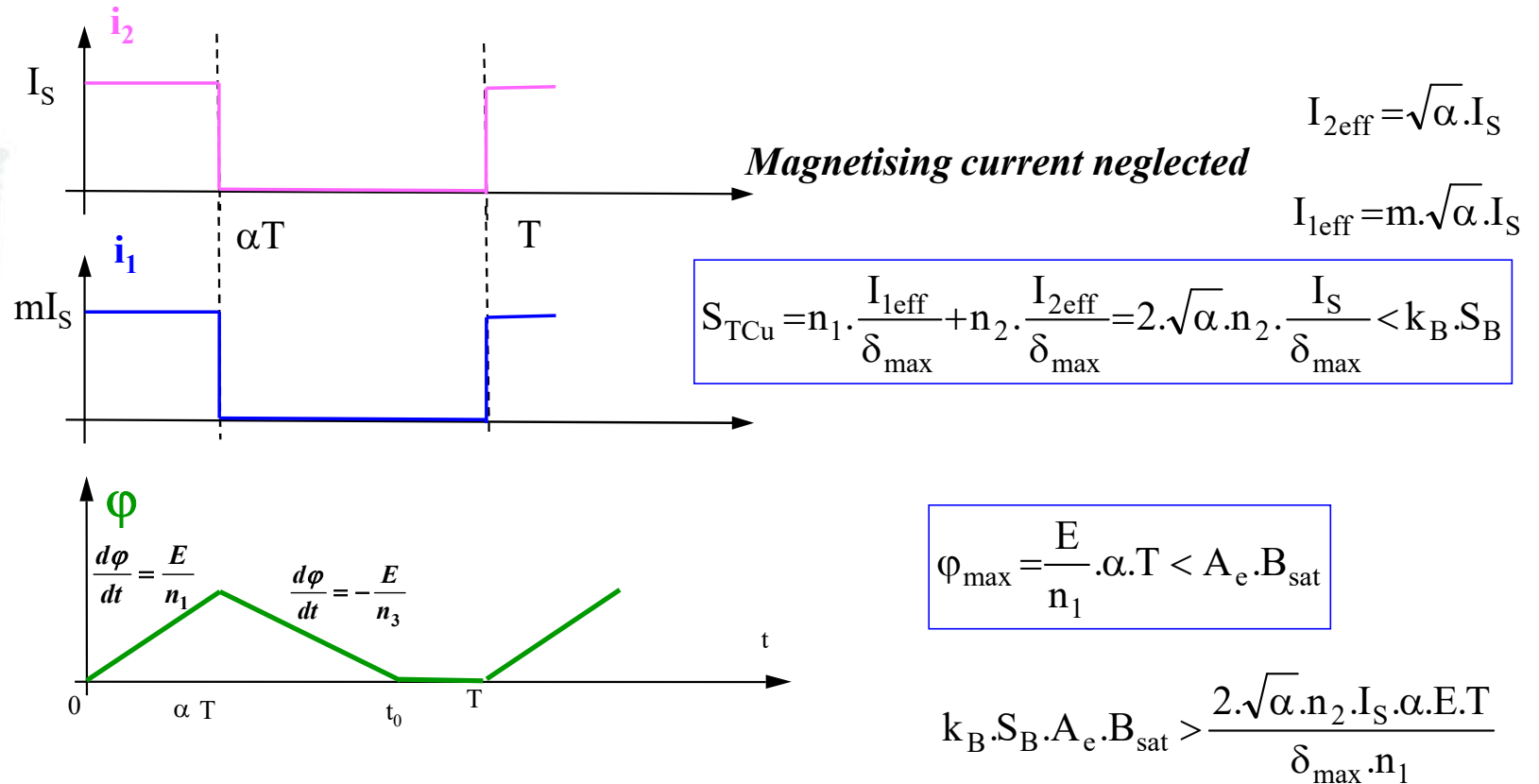
Flyback

Forward

Principle
Continuous mode
Sizing

Push-Pull

Continuous mode study: transformer sizing



$$A_e \cdot S_B > \frac{2 \cdot \sqrt{\alpha} \cdot P_S}{f \cdot k_B \cdot B_{sat} \cdot \delta_{max}}$$

With P_S

With f

Introduction

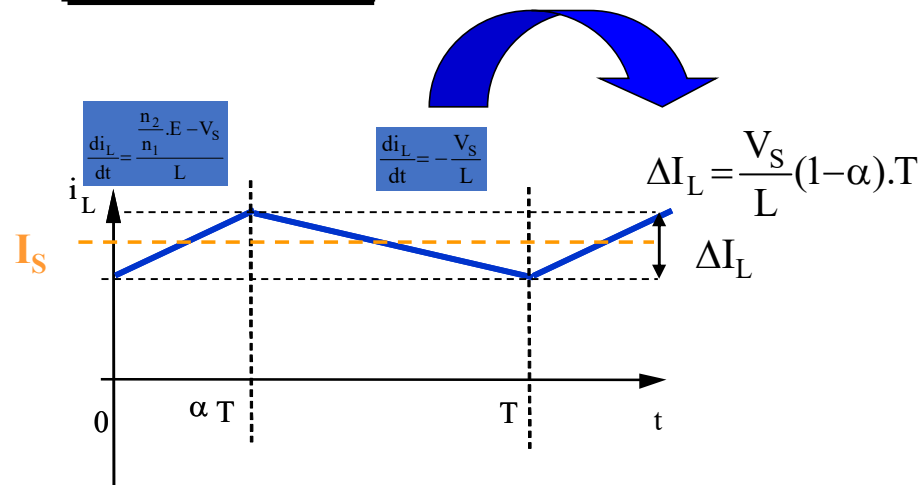
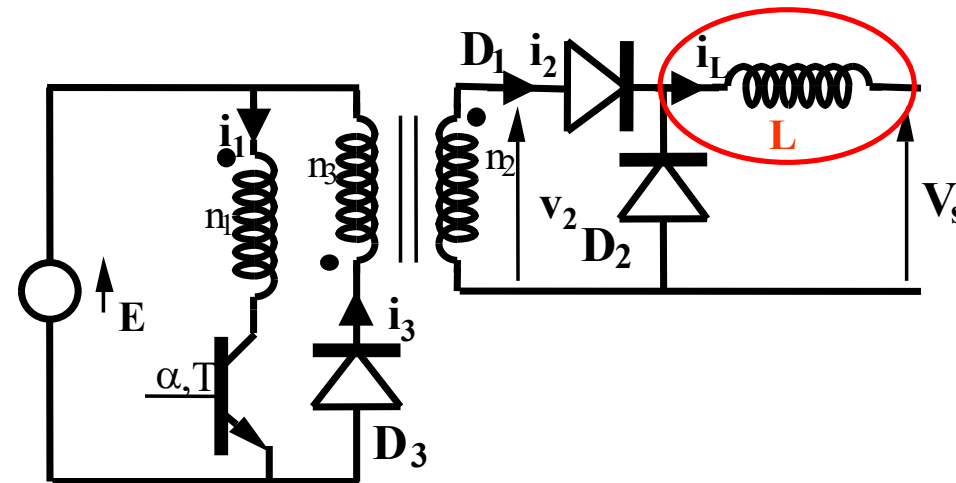
Flyback

Forward

Principle
Continuous mode
Sizing

Push-Pull

Continuous mode study: inductor sizing



Hold the continuous mode

$$I_s > \frac{\Delta I_L}{2}$$

$$L > \frac{\alpha \cdot V_s}{2 I_s \cdot f}$$

Introduction

Flyback

Forward

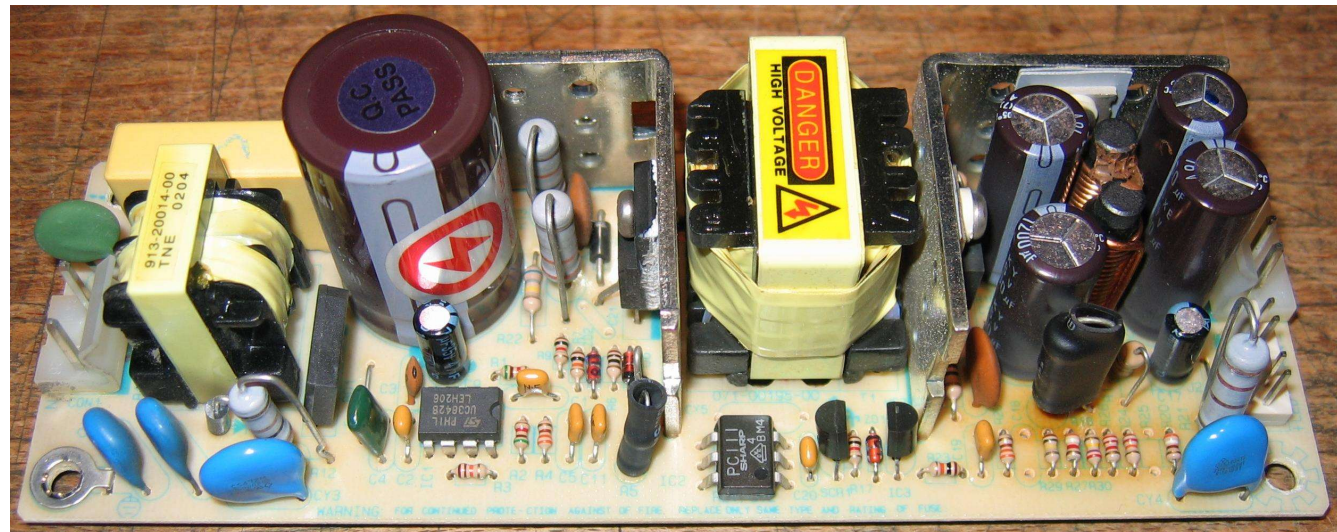
Principle

Continuous mode

Sizing

Push-Pull

Example



Manufacturer reference: CI Design Y0-2040AA

Input : 100 à 240 Vac

Output : 12 Vdc (2 A) 5 Vdc (3 A)

Introduction

Flyback

Forward

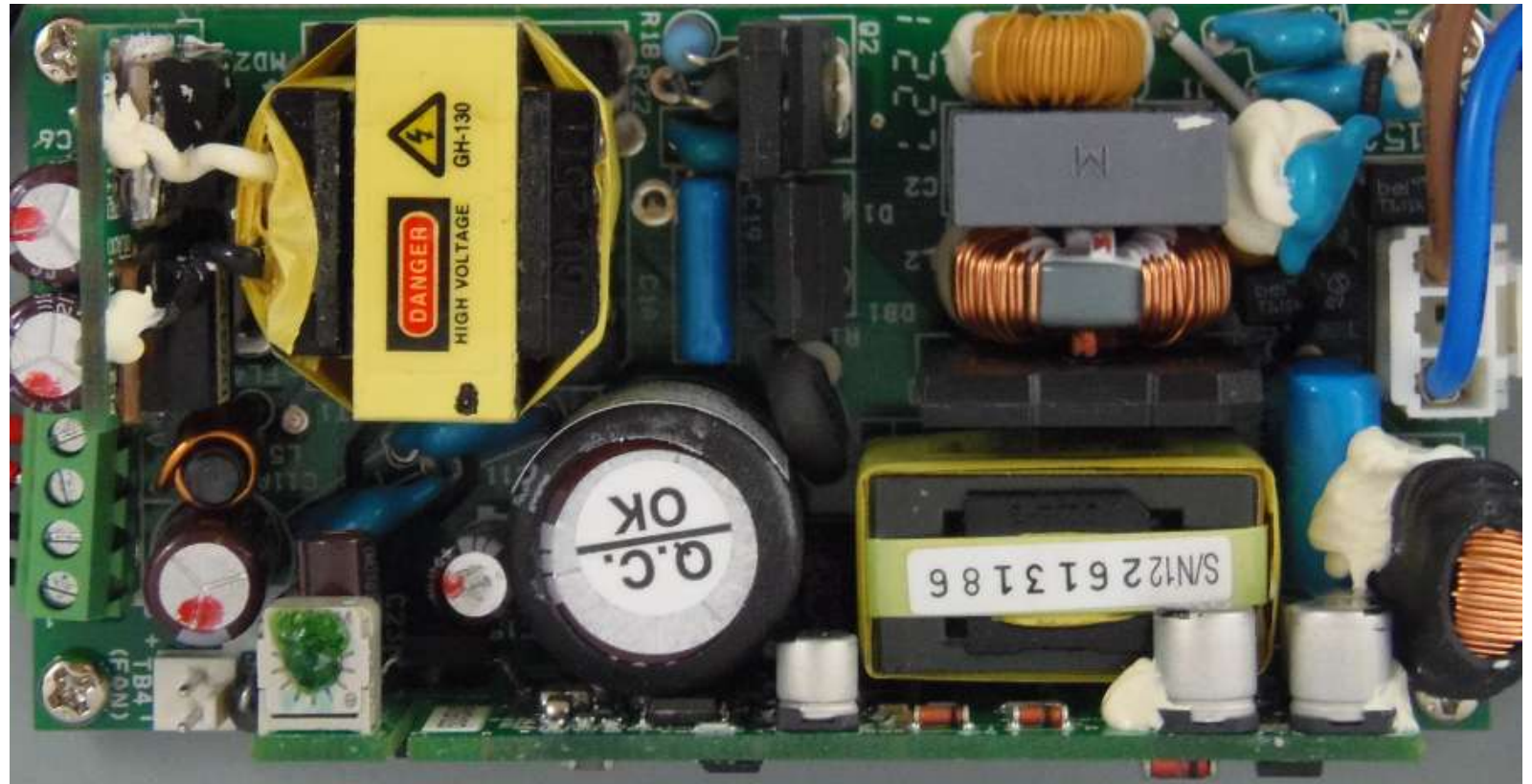
Principle

Continuous mode

Sizing

Push-Pull

Example



NuForce

Introduction

Flyback

Forward

Push-Pull

Push-Pull

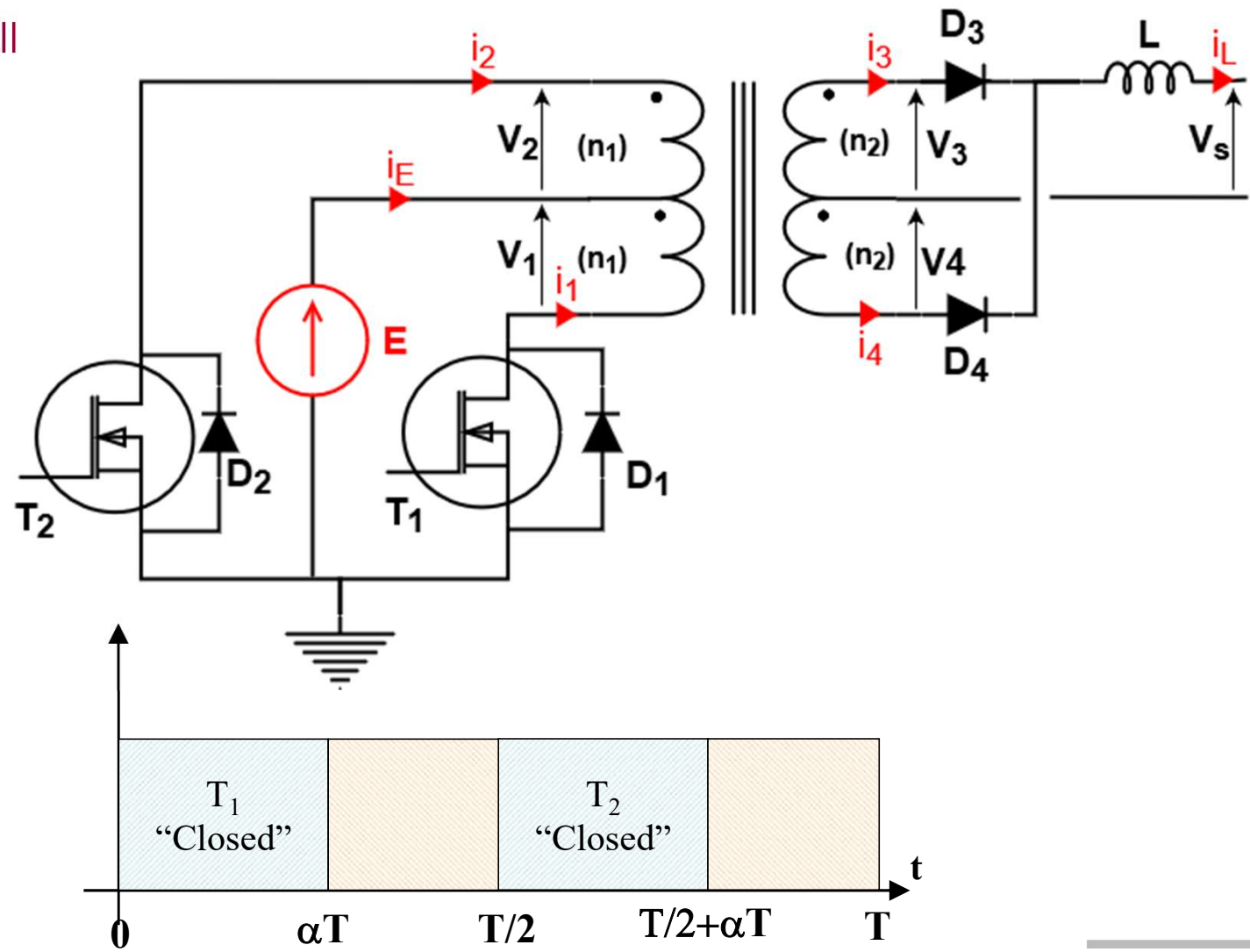
Introduction

Flyback

Forward

Push-Pull

Push-Pull



Introduction

Flyback

Forward

Push-Pull

No load study

Continuous mode

Other structures

No load study : $i_L = 0$

$0 < t < \alpha T$: T_1 "closed", T_2 "open"

- $v_1 = n_1 \cdot \frac{d\varphi}{dt} = E > 0$

Magnetization

- $v_2 = v_1 = E > 0$

➡ **D_2 « reverse »**

$$-n_1 \cdot i_1 = \mathcal{R} \cdot \varphi$$

$\alpha T < t < T/2$: T_1 & T_2 "open"

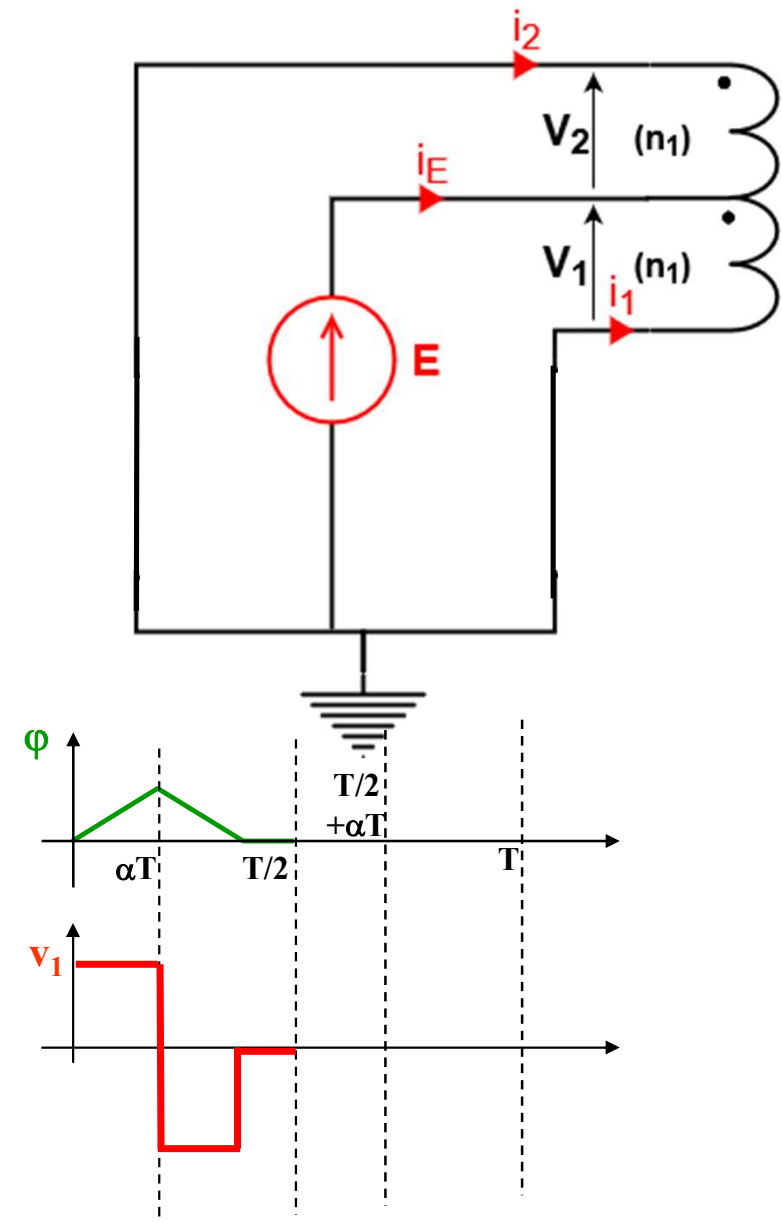
- $\mathcal{R} \cdot \varphi = -n_1 \cdot i_1 + n_1 \cdot i_2 > 0$

➡ **D_2 « forward », D_1 « reverse »**

- $\mathcal{R} \cdot \varphi = n_1 \cdot i_2 > 0$

- $v_2 = n_1 \cdot \frac{d\varphi}{dt} = -E < 0$

Demagnetization



Introduction

Flyback

Forward

Push-Pull

No load study

Continuous mode

Other structures

No load study : $i_L = 0$

$T/2 < t < T/2 + \alpha T$: T_1 "open", T_2 "closed"

- $v_1 = n_1 \cdot \frac{d\varphi}{dt} = -E < 0$

Magnetization

- $v_2 = v_1 = -E < 0$

➡ **D_1 « reverse »**

$$n_1 \cdot i_2 = \mathcal{R} \cdot \varphi$$

$T/2 + \alpha T < t < T$: T_1 & T_2 "open"

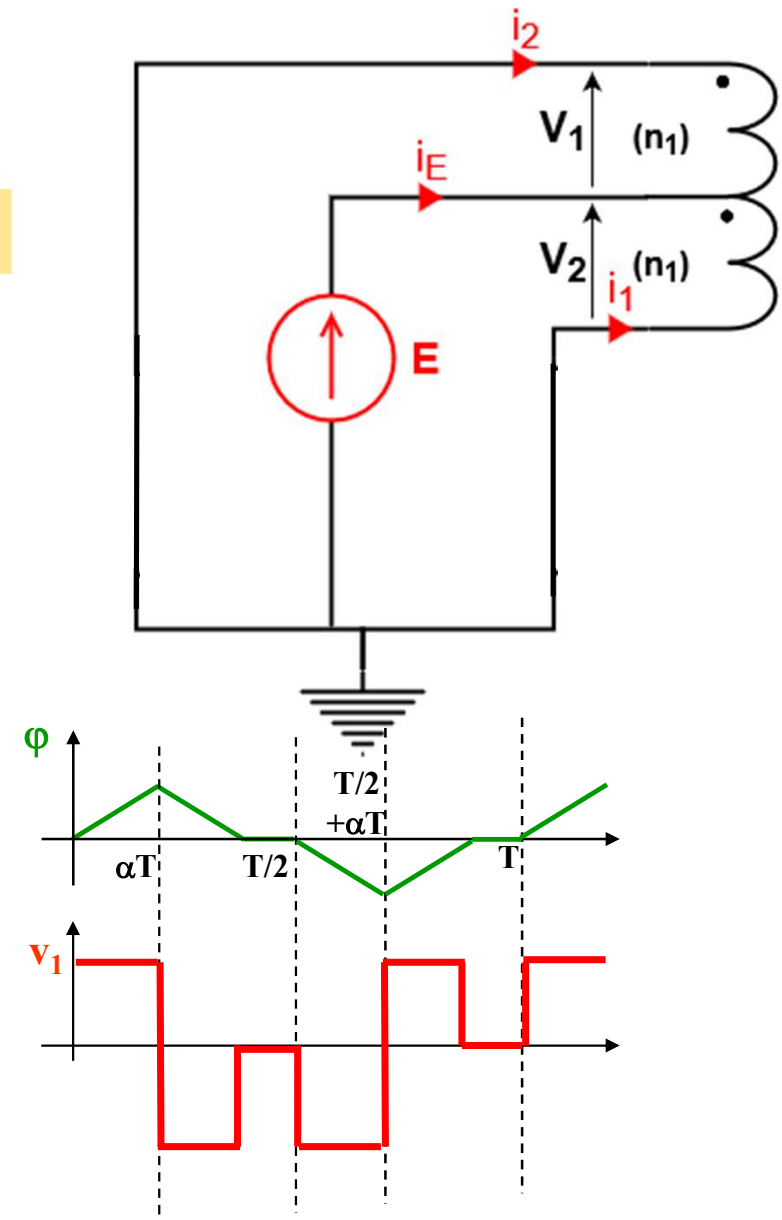
- $\mathcal{R} \cdot \varphi = -n_1 \cdot i_1 + n_1 \cdot i_2 < 0$

➡ **D_1 « forward », D_2 « reverse »**

- $\mathcal{R} \cdot \varphi = -n_1 \cdot i_1 < 0$

- $v_2 = n_1 \cdot \frac{d\varphi}{dt} = E > 0$

Demagnetization



Introduction

Flyback

Forward

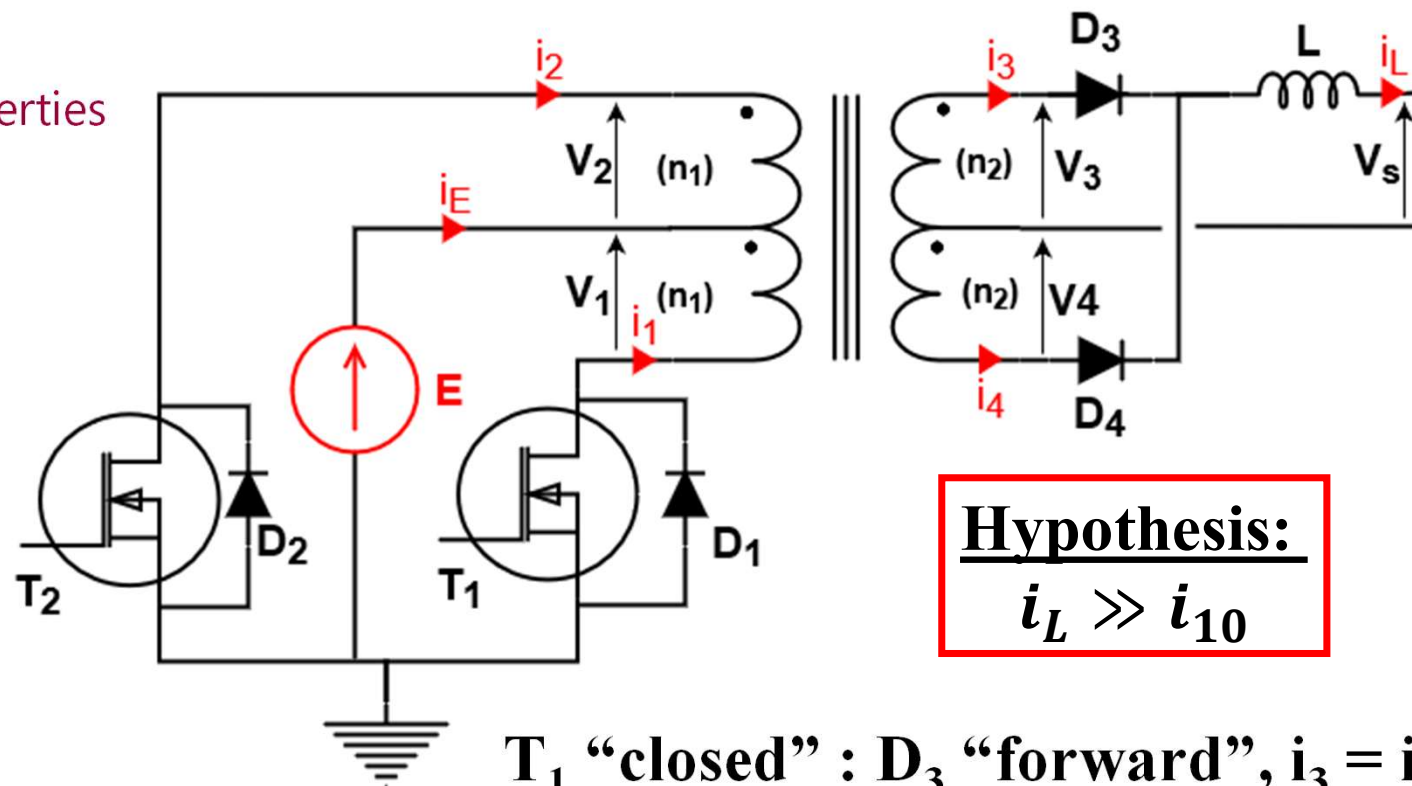
Push-Pull

No load study

Continuous mode

Other structures

General properties



Hypothesis:

$$i_L \gg i_{10}$$

$$\begin{aligned} v_1 &= v_2 \\ v_3 &= v_4 \\ v_3 &= \frac{n_2}{n_1} v_1 \end{aligned}$$

T₁ “closed” : D₃ “forward”, i₃ = i_L

T₂ “closed” : D₄ “forward”, i₄ = i_L

Hopkinson's law

$$\oint H \cdot dl = \mathcal{R} \cdot \varphi = -n_1 \cdot i_1 + n_1 \cdot i_2 - n_2 \cdot i_3 + n_2 \cdot i_4$$

Introduction

Flyback

Forward

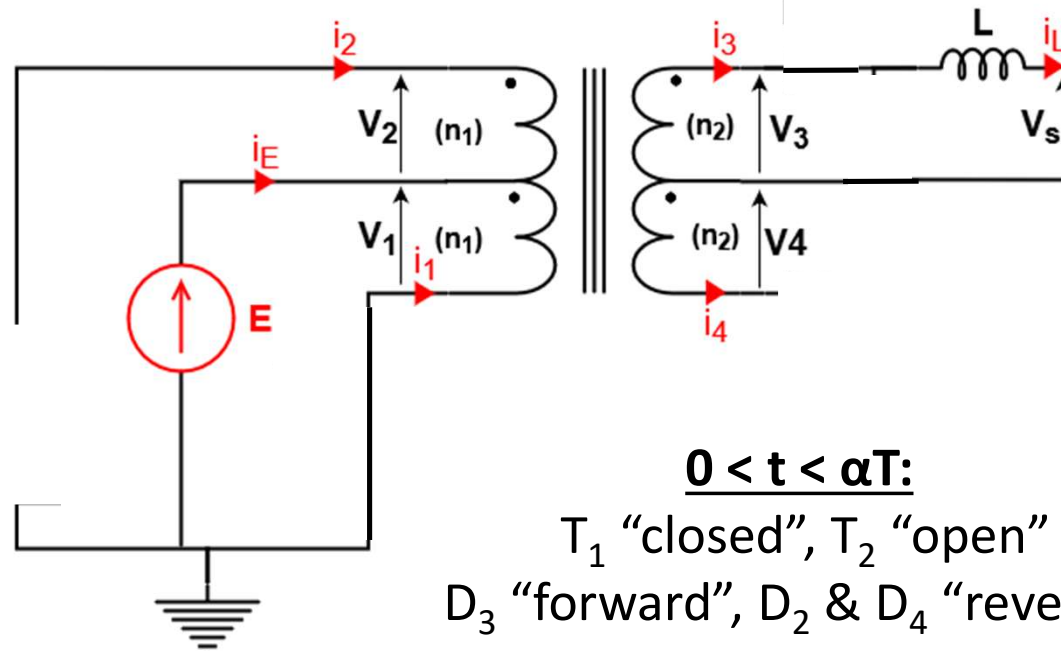
Push-Pull

No load study

Continuous mode

Other structures

Continuous mode study: magnetization



$0 < t < \alpha T$:

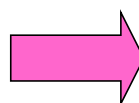
T_1 "closed", T_2 "open"

D_3 "forward", D_2 & D_4 "reverse"

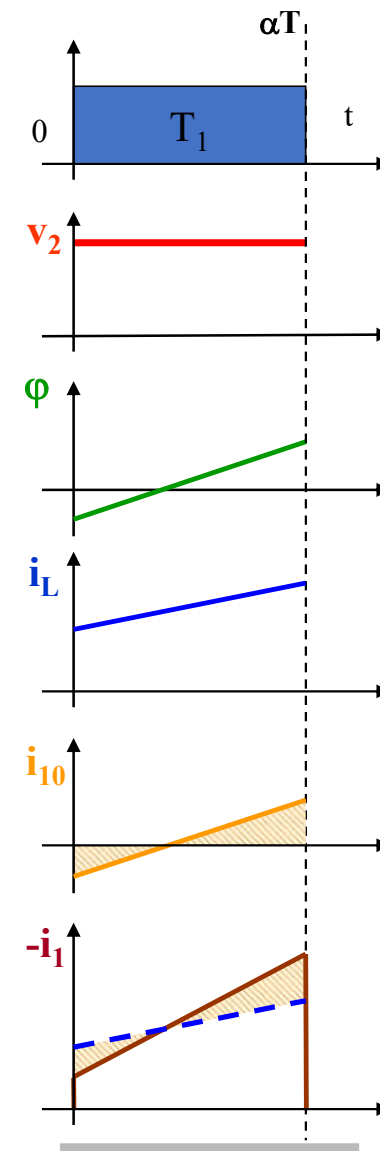
- $v_1 = n_1 \cdot \frac{d\phi}{dt} = E > 0$

Magnetization

$$-n_1 \cdot i_1 - n_2 \cdot i_3 = \mathcal{R} \cdot \phi$$



$$i_1 = -\frac{\mathcal{R}}{n_1} \cdot \phi - \frac{n_2}{n_1} \cdot i_3 = -i_{10} - \frac{n_2}{n_1} \cdot i_L$$



Introduction

Flyback

Forward

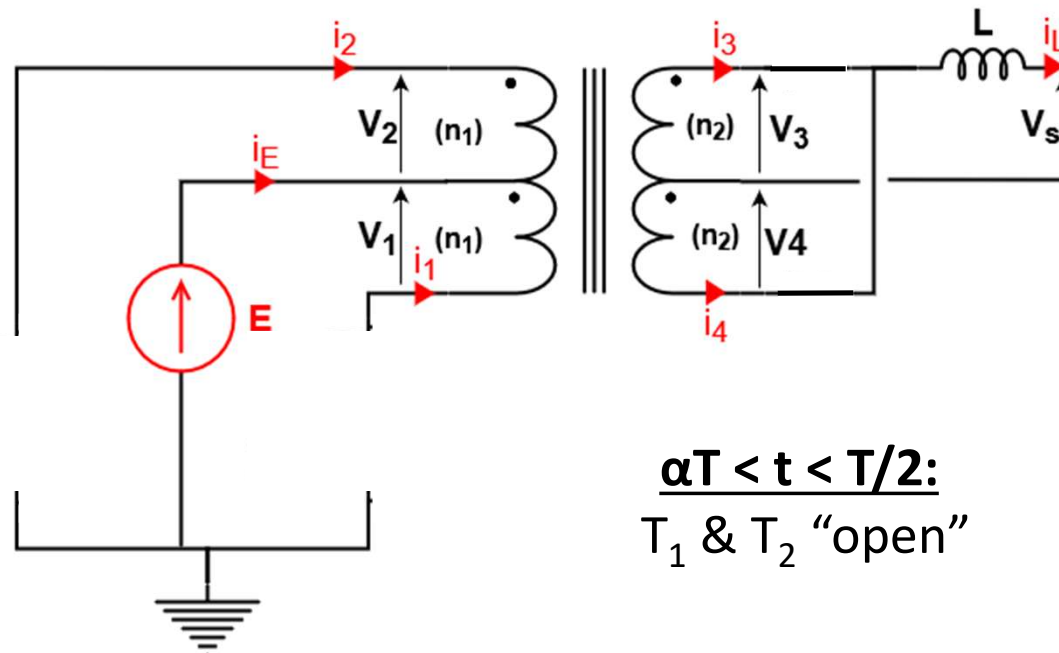
Push-Pull

No load study

Continuous mode

Other structures

Continuous mode study: second phase



$\alpha T < t < T/2$:
 T_1 & T_2 "open"

$$\mathcal{R}.\varphi = -n_1.i_1 + n_1.i_2 - n_2.i_3 + n_2.i_4 > 0$$

- If D_2 « forward » $\rightarrow D_4$ « forward »

$$\mathcal{R}.\varphi = n_1.i_2 + n_2.i_L$$

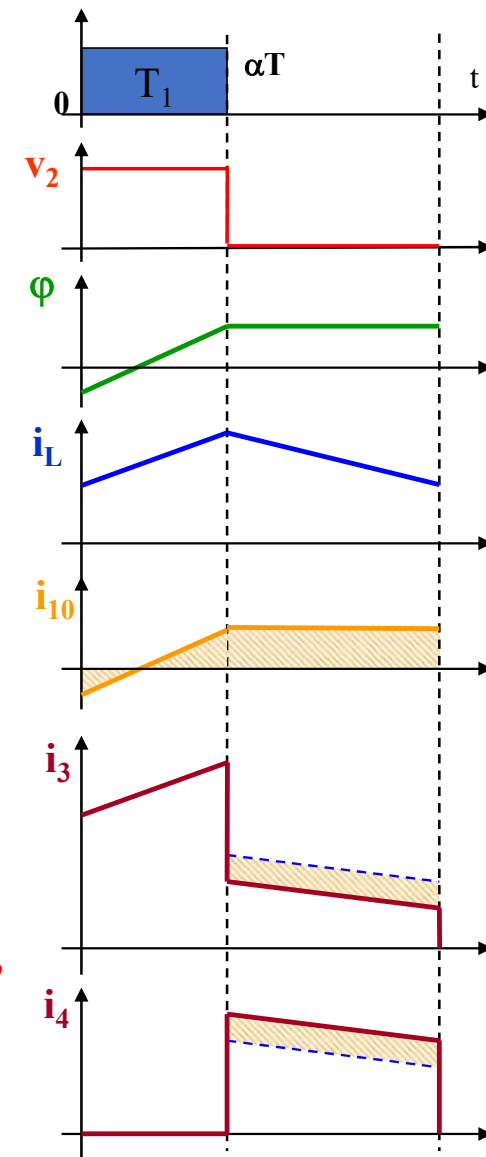
$$\mathcal{R}.\varphi = -n_2.i_3 + n_2.i_4 > 0$$

$$v_2 = v_3 = v_4 = 0$$

$\Rightarrow D_2$ "reverse"

$\Rightarrow D_3$ & D_4 "forward"

$\Rightarrow \varphi = cst$



Introduction

Flyback

Forward

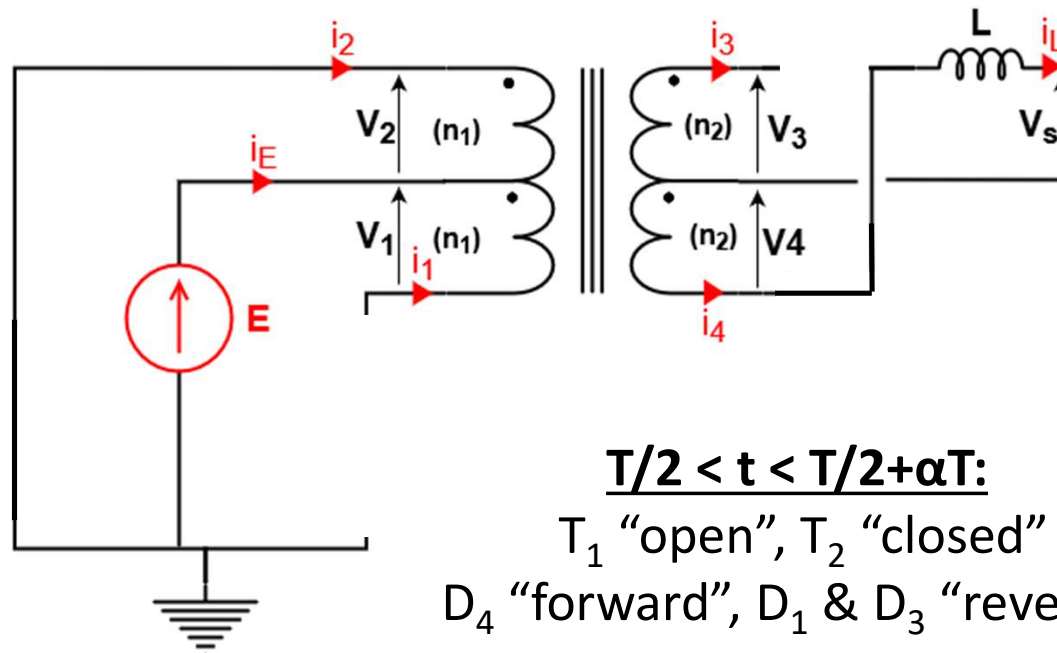
Push-Pull

No load study

Continuous mode

Other structures

Continuous mode study: demagnetization



$T/2 < t < T/2 + \alpha T$

T_1 "open", T_2 "closed"

D_4 "forward", D_1 & D_3 "reverse"

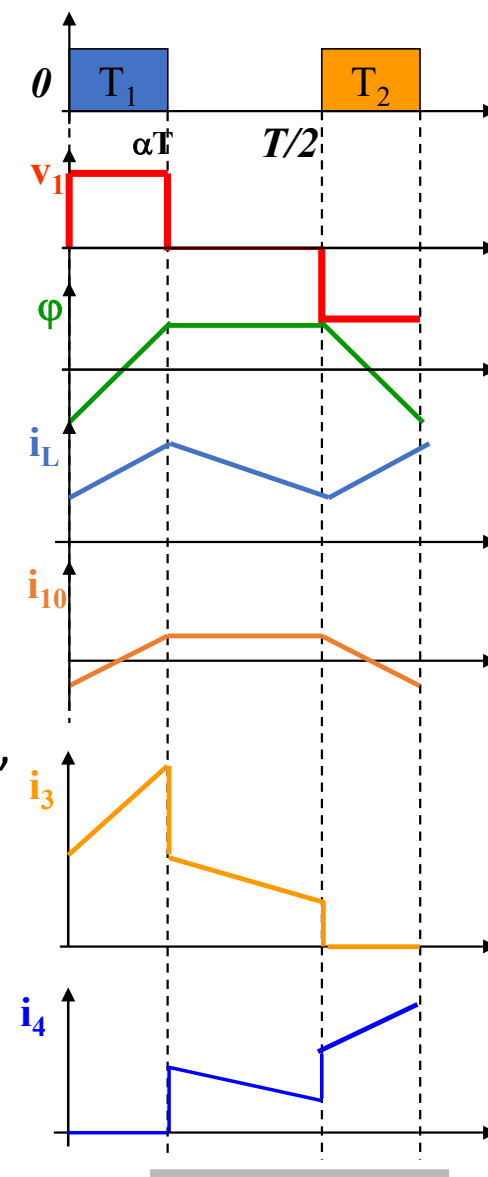
- $v_2 = n_1 \cdot \frac{d\phi}{dt} = -E < 0$

Demagnetization

$$n_1 \cdot i_2 + n_2 \cdot i_4 = \mathcal{R} \cdot \phi$$



$$i_2 = \frac{\mathcal{R}}{n_1} \cdot \phi - \frac{n_2}{n_1} \cdot i_4 = i_{10} - \frac{n_2}{n_1} \cdot i_L$$



Introduction

Flyback

Forward

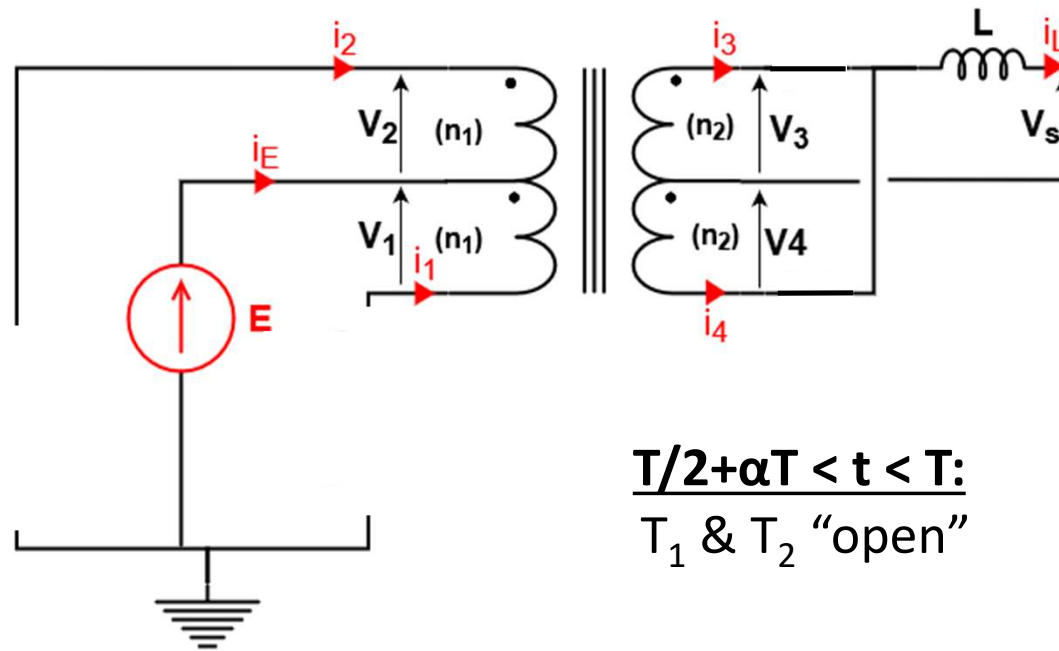
Push-Pull

No load study

Continuous mode

Other structures

Continuous mode study: fourth phase



$\frac{T}{2} + \alpha T < t < T$
 $T_1 \text{ \& } T_2 \text{ "open"}$

$$\mathcal{R}.\varphi = -n_1.i_1 + n_1.i_2 - n_2.i_3 + n_2.i_4 < 0$$

- If $D_1 \ll \text{forward}$

$$\mathcal{R}.\varphi = -n_1.i_1 - n_2.i_L$$

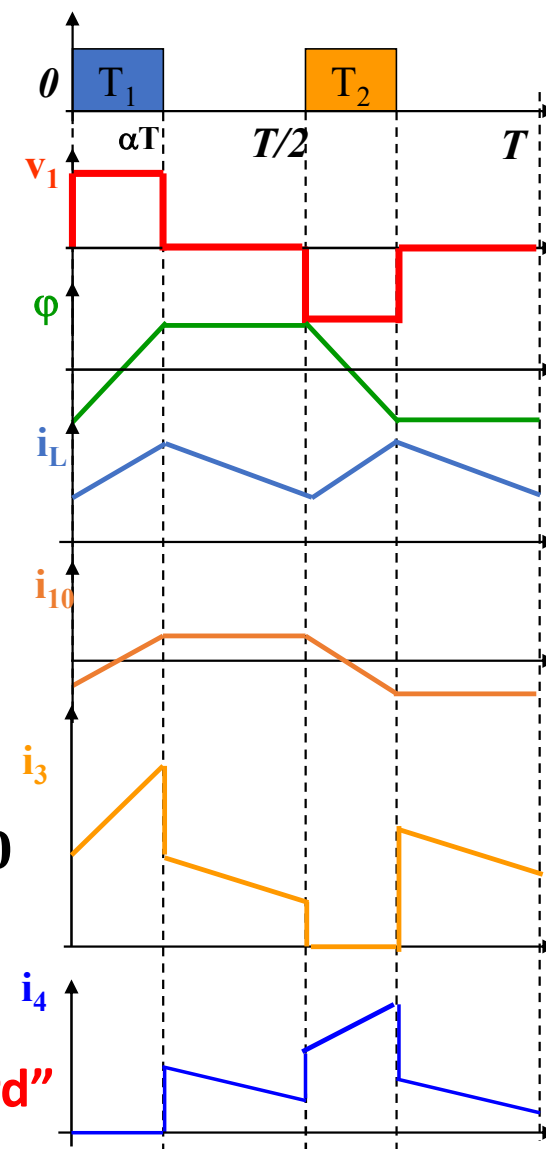
$$\mathcal{R}.\varphi = -n_2.i_3 + n_2.i_4 < 0$$

$$v_2 = v_3 = v_4 = 0$$

D_1 "reverse"

$D_3 \text{ \& } D_4$ "forward"

$\varphi = cst$



Introduction

Flyback

Forward

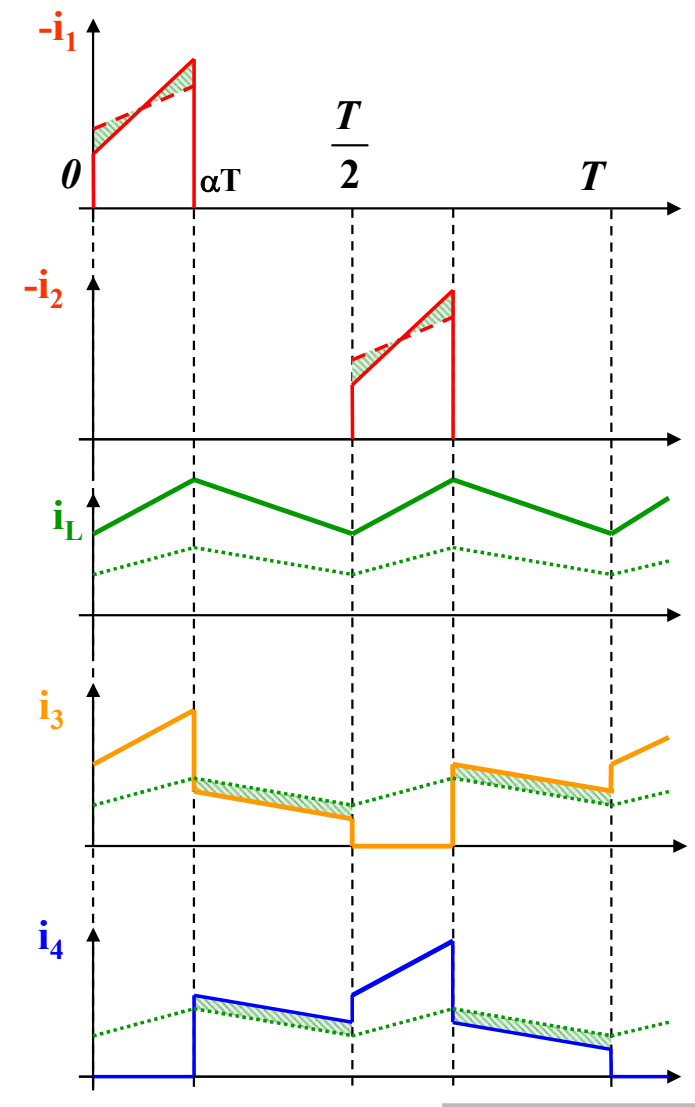
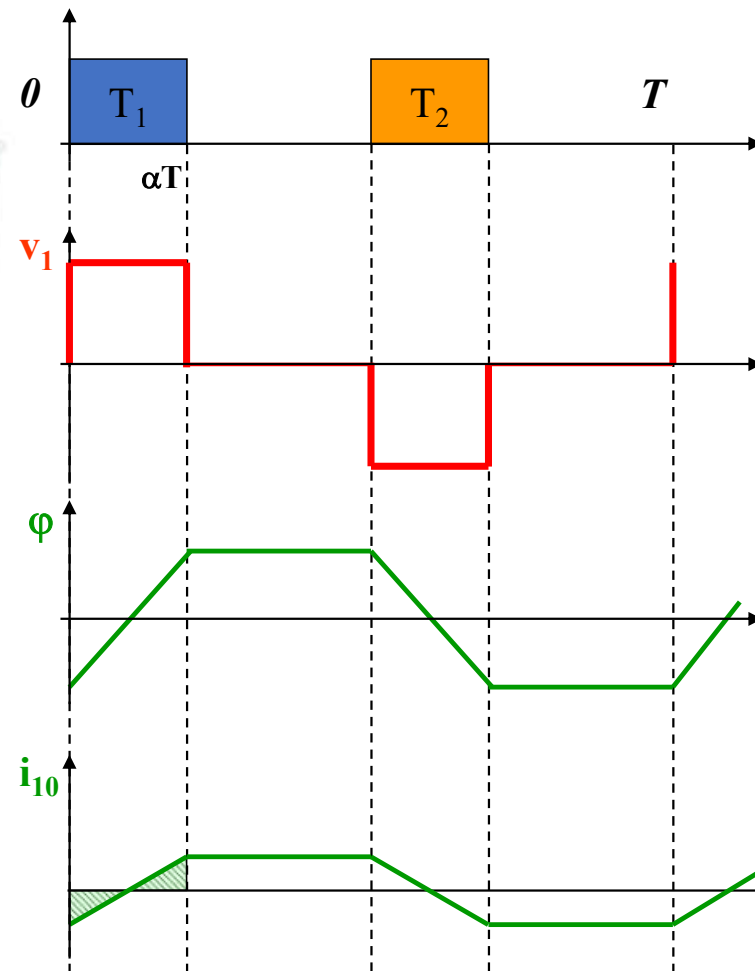
Push-Pull

No load study

Continuous mode

Other structures

Continuous mode study: chronograms



Introduction

Flyback

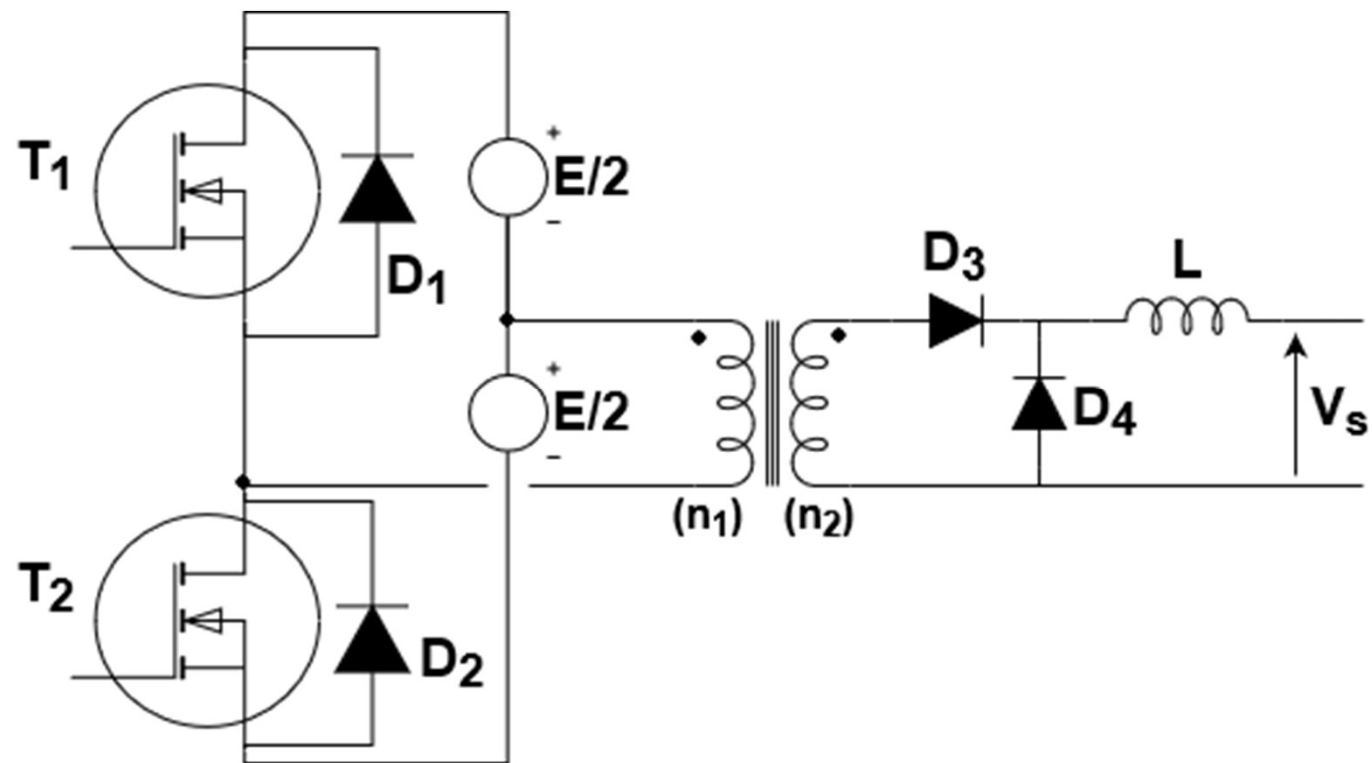
Forward

Push-Pull

No load study
Continuous mode

Other structures

Other symmetrical structures : Half Bridge Forward



- Same behavior as Push-Pull
- Only on primary and secondary winding
- Transformer simpler (so cheaper)

Introduction

Flyback

Forward

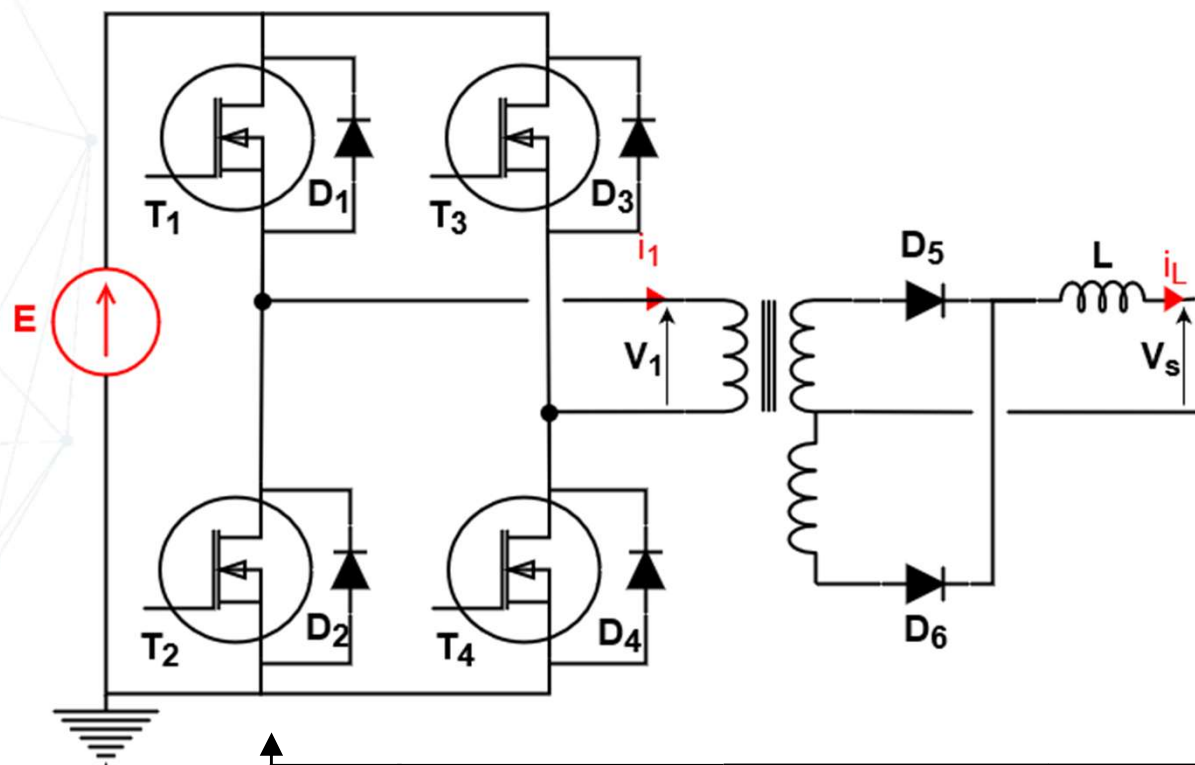
Push-Pull

No load study

Continuous mode

Other structures

Other symmetrical structures: Full Bridge Forward



- Same behavior with no-load and full load
- Control with the shift between the two arms

